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Chapter #5: MOSFET's

from **Microelectronic Circuits** Text

by Sedra and Smith

Oxford Publishing

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Introduction

- **IN THIS CHAPTER WE WILL LEARN**
  - The physical structure of the MOS transistor and how it works.
  - How the voltage between two terminals of the transistor control the current that flows through the third terminal, and the equations that describe these current-voltage characteristics.
  - How the transistor can be used to make an amplifier, and how it can be used as a switch in digital circuits.

Introduction

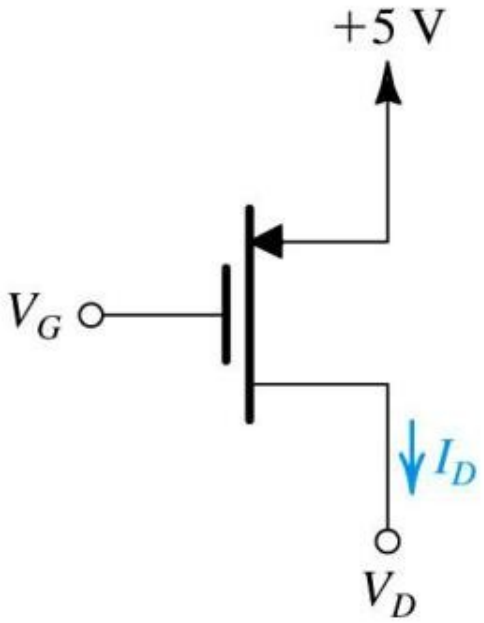
- **IN THIS CHAPTER WE WILL LEARN**
  - How to obtain linear amplification from the fundamentally nonlinear MOS transistor.
  - The three basic ways for connecting a MOSFET to construct amplifiers with different properties.
  - Practical circuits for MOS-transistor amplifiers that can be constructed using discrete components.

Introduction

- We have studied two-terminal semi-conductor devices (e.g. diode).
- However, now we turn our attention to three-terminal devices.
- They are more useful because they present multitude of applications, e.g:
  - signal amplification, digital logic, memory, etc…

Introduction

- **Q:** What, in simplest terms, is the desired operation of a three-terminal device?
  - **A:** Employ voltage between two terminals to control current flowing in to the third.



Introduction

note: MOSFET is more widely used in implementation of modern electronic devices

- **Q:** What are two major types of three-terminal semiconductor devices?
  - metal-oxide-semiconductor field-effect transistor (MOSFET)
  - bipolar junction transistor (BJT)
- **Q:** Why are MOSFET's more widely used?
  - size (smaller)
  - ease of manufacture
  - lesser power utilization

- MOSFET technology
  - It allows placement of approximately 2 billion transistors on a single IC
    - backbone of very large scale integration (VLSI)
  - It is considered preferable to BJT technology for many applications.

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5.1. **Device Structure and Operation**

- Figure 5.1. shows general structure of the *n*-channel enhancement-type MOSFET

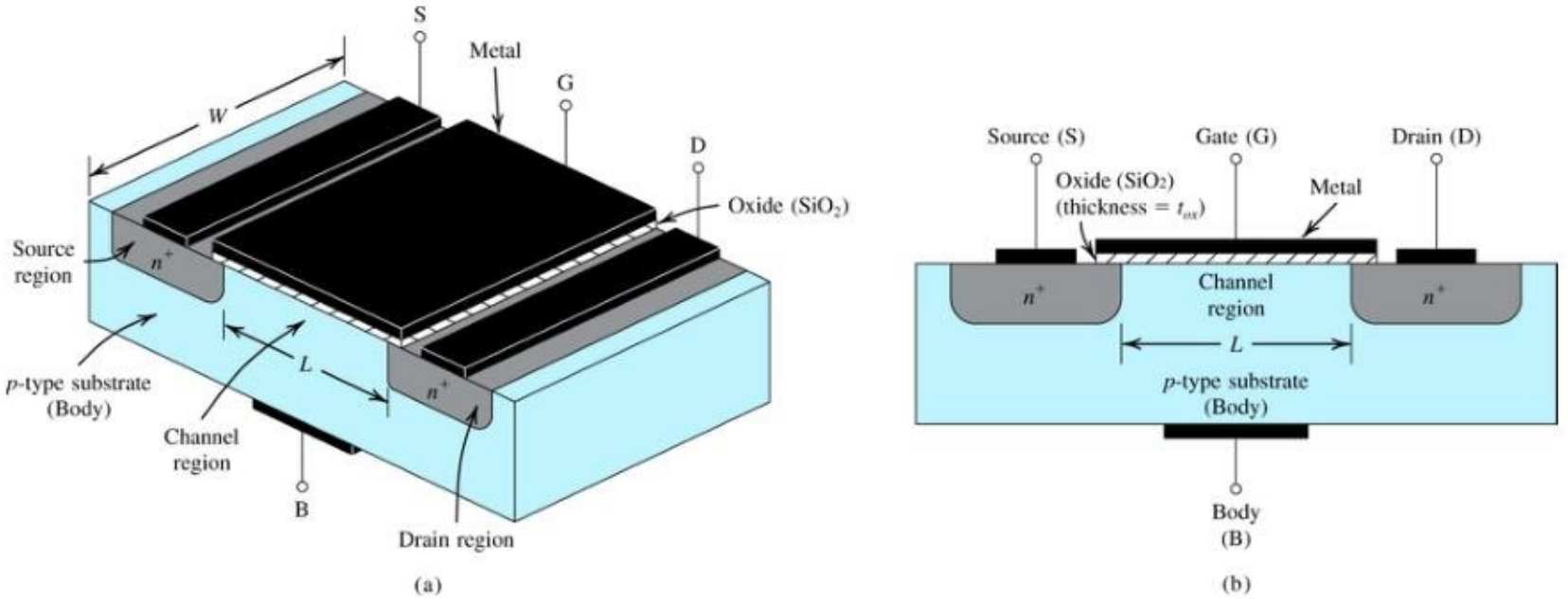


Figure 5.1: Physical structure of the enhancement-type NMOS transistor: **(a)** perspective view, **(b)** cross-section. Note that typically L = 0.03*um* to 1*um*, W = 0.1*um* to 100*um*, and the thickness of the oxide layer (t_{ox}) is in the range of 1 to 10*nm*.

5.1. **Device Structure and Operation**

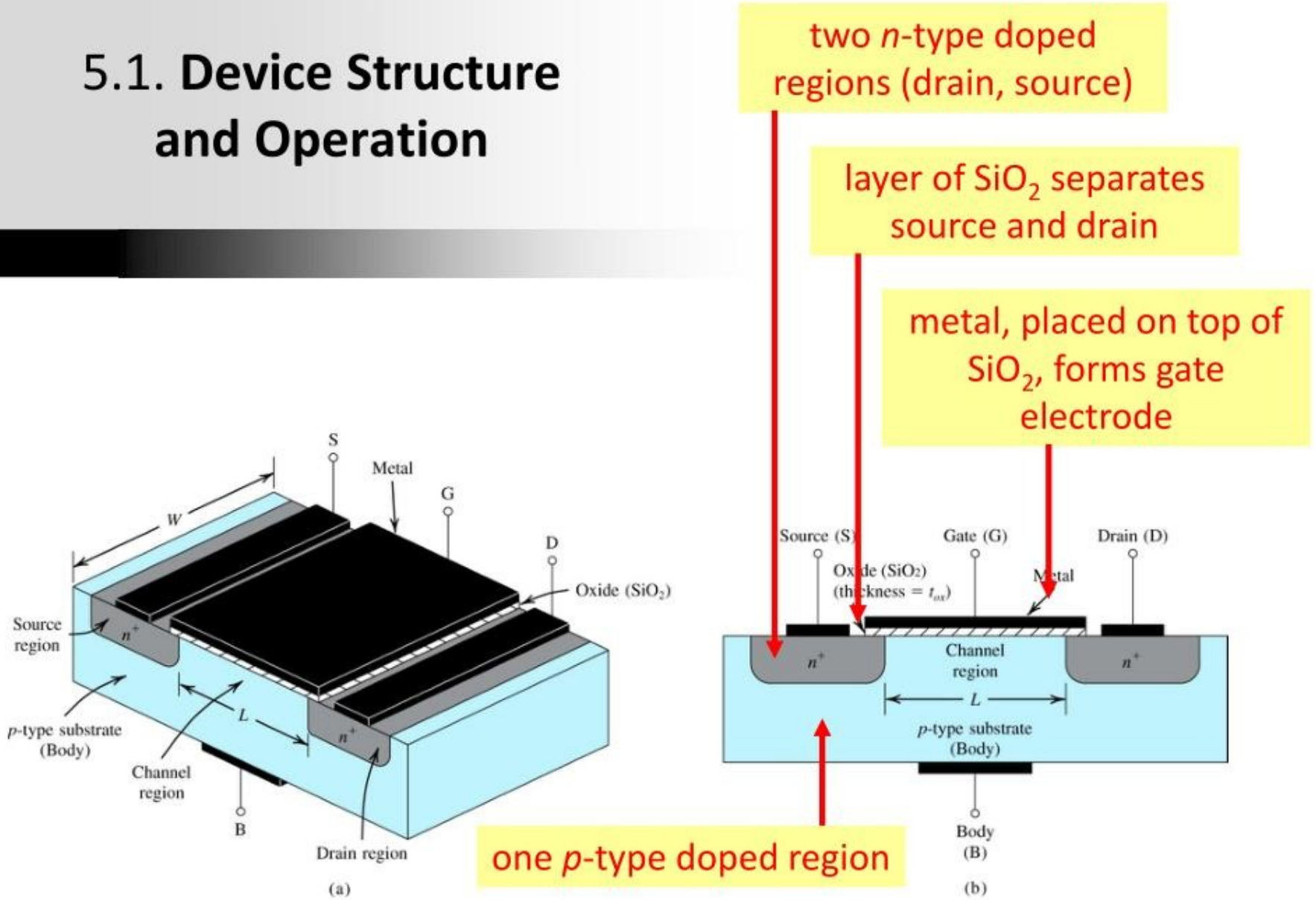


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5.1. **Device Structure and Operation**

- The name MOSFET is derived from its physical structure.
- However, many MOSFET's do not actually use any "metal", polysilicon is used instead.
  - "This" has no effect on modeling / operation as described here.
- Another name for MOSFET is insulated gate FET, or IGFET.
- The device is composed of two pn-junctions, however they maintain reverse biasing at all times.
  - Drain will always be at positive voltage with respect to source.
- We will not consider conduction of current in this manner.

5.1.2. **Operation with Zero Gate Voltage**

- With zero voltage applied to gate, two back-to-back diodes exist in series between drain and source.
- "They" prevent current conduction from drain to source when a voltage v_{DS} is applied.
  - yielding very high resistance (10^{12}*ohms*)

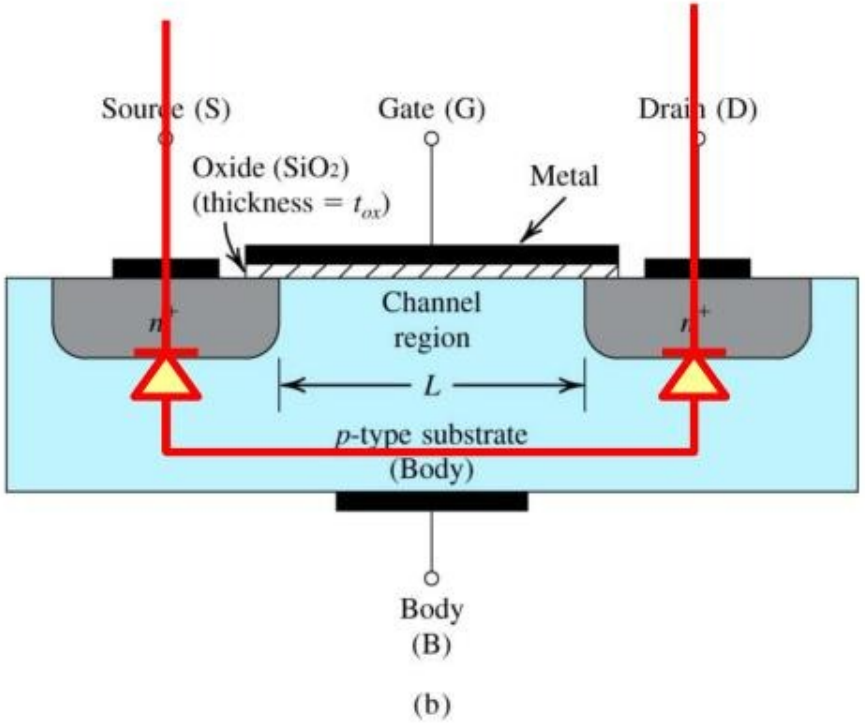


Figure 5.1: Physical structure...

5.1.3. Creating a Channel for Current Flow

- **Q:** What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.
  - **step #1:** v_{GS} is applied to the gate terminal, causing a positive build up of positive charge along metal electrode.
  - **step #2:** This "build up" causes free holes to be repelled from region of *p*-type substrate under gate.

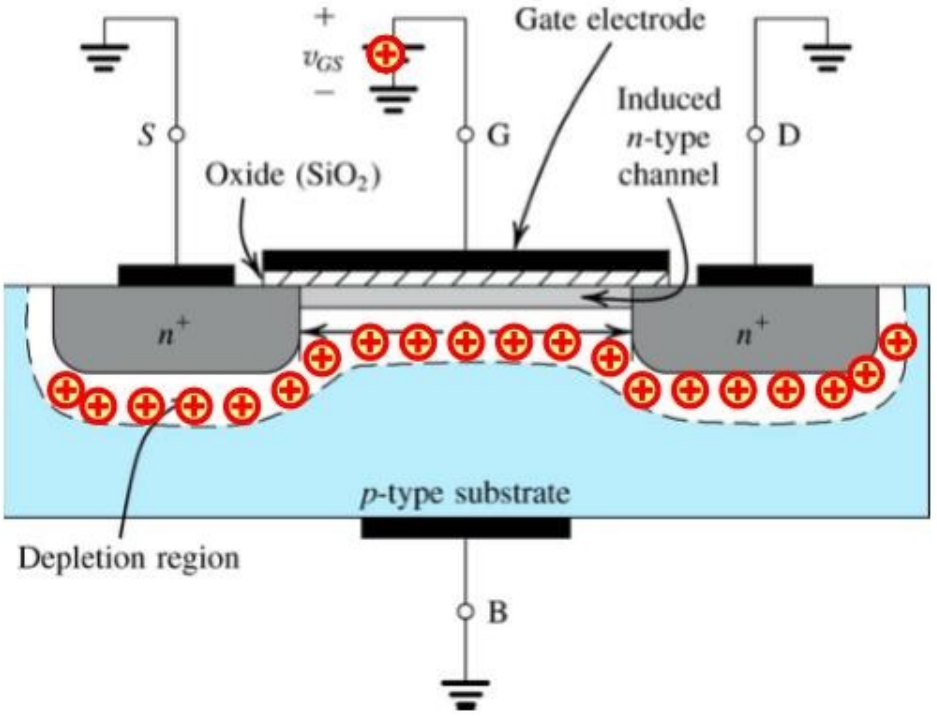


Figure 5.2: The enhancement-type NMOS transistor with a positive voltage applied to the gate. An *n* channel is induced at the top of the substrate beneath the gate

Q: What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.

- **step #3:** This "migration" results in the uncovering of negative **bound charges**, originally neutralized by the free holes
- **step #4:** The positive gate voltage also attracts electrons from the n^+ source and drain regions into the channel.

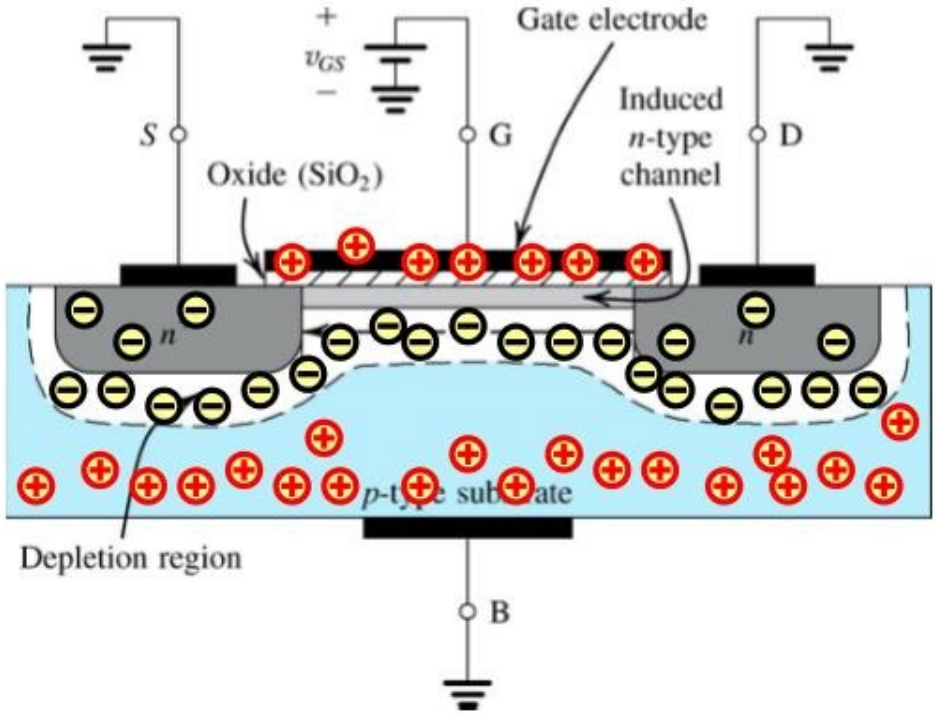


Figure 5.2: The enhancement-type NMOS transistor with a positive voltage applied to the gate. An *n* channel is induced at the top of the substrate beneath the gate

Q: What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.

this induced channel is also known as an **inversion layer**

- **step #5:** Once a sufficient number of "these" electrons accumulate, an *n*-region is created…
  - …connecting the source and drain regions
- **step #6:** This provides path for current flow between *D* and *S*.

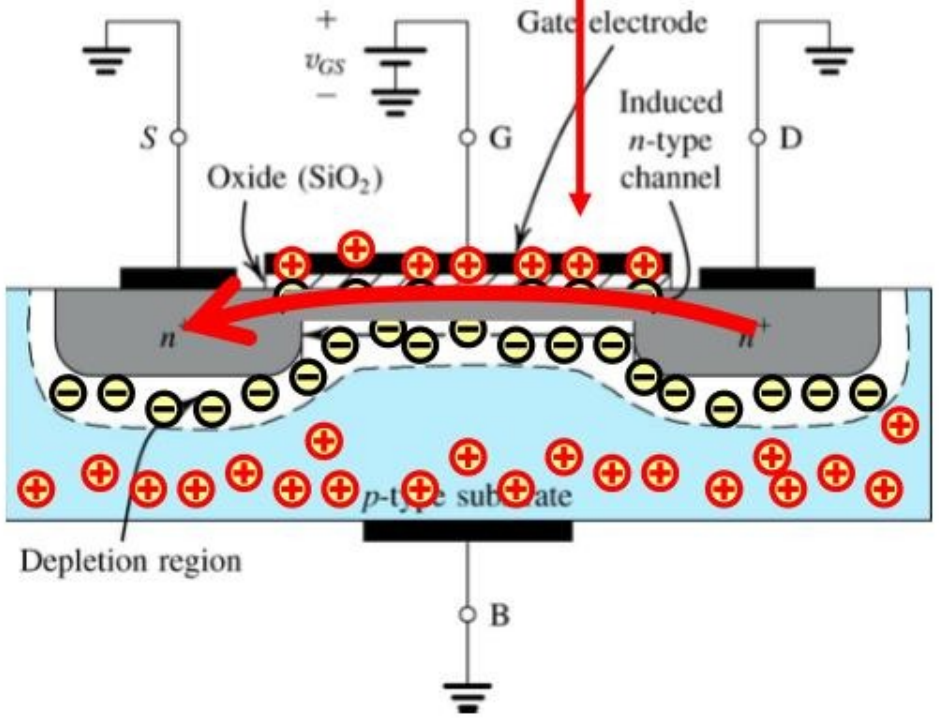


Figure 5.2: The enhancement-type NMOS transistor with a positive voltage applied to the gate. An *n* channel is induced at the top of the substrate beneath the gate

5.1.3. Creating a Channel for Current Flow

V_{tn} is used for *n*-type MOSFET, V_{tp} is used for *p*-channel

- **threshold voltage** (V_t) – is the minimum value of v_{GS} required to form a conducting channel between drain and source
  - typically between 0.3 and 0.6V_{dc}
- **field-effect** – when positive v_{GS} is applied, an electric field develops between the gate electrode and induced *n*-channel – the conductivity of this channel is affected by the strength of field
  - SiO_2 layer acts as dielectric

- **effective / overdrive voltage** – is the difference between v_{GS} applied and V_t.

$$(eq5.1)\quad v_{OV} \equiv v_{GS} - V_t$$

- **oxide capacitance** (C_{ox}) – is the capacitance of the parallel plate capacitor per unit gate area (F/m^2)

ε_{ox} is permittivity of $SiO_2 = 3.45\mathbf{E}{-}11(F/m)$
t_{ox} is thickness of SiO_2 layer

$$(eq5.3)\quad C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} \text{ in } F/m^2$$

5.1.3. Creating a Channel for Current Flow

- **Q:** What is main requirement for *n*-channel to form?
  - **A:** The voltage across the "oxide" layer must exceed V_t.
- For example, when $v_{DS} = 0$...
  - the voltage at every point along channel is zero
  - the voltage across the oxide layer is uniform and equal to v_{GS}

- **Q:** How can one express the magnitude of electron charge contained in the channel?
  - **A:** See below...

W and L represent width and length of channel respectively

(eq5.2) $|Q| = C_{ox}(WL)v_{OV}$ in C

- **Q:** What is effect of v_{OV} on *n*-channel?
  - **A:** As v_{OV} grows, so does the depth of the *n*-channel as well as its conductivity.

5.1.4. **Applying a Small v_{DS}**

- **Q:** For small values of v_{DS}, how does one calculate i_{DS} (aka. i_D)? **A:** Equation (5.7)…
- **Q:** What is the origin of this equation?
  - **A:** Current is defined in terms of **charge per unit length of *n*-channel** as well as **electron drift velocity.**

μ_n represents mobility of electrons at surface of the n-channel in m^2/Vs

$$\text{(eq5.7)}\ i_D = \underbrace{\left(C_{ox} W v_{OV}\right)}_{\text{charge per unit length of } n\text{-channel in } C/m} \underbrace{\left(\frac{\mu_n v_{DS}}{L}\right)}_{\text{electron drift velocity in } m^2/Vs} \text{ in } A$$

5.1.4. **Applying a Small v_{DS}**

- **Q:** How does one calculate charge per unit length of *n*-channel (Q/uL)?
  - **A:** For small values of v_{DS}, one can still assume that voltage between gate and *n*-channel is constant (along its length) – and equal to v_{GS}.
  - **A:** Therefore, effective voltage between gate and *n*-channel remains equal to v_{OV}.
  - **A:** Therefore, (5.2) from two slides back applies.

5.1.4. **Applying a Small v_{DS}**

- **Q:** How does one calculate charge per unit length of *n*-channel (Q/uL)?
  - **A:** Use (5.2) to calculate charge per unit L of channel.
- **Q:** How does one calculate electron drift velocity?
  - **A:** Note that v_{DS} establishes an electric field E across length of *n*-channel, this may calculate e-drift velocity.

action: divide both sides by L

(eq5.2) $|Q| = C_{ox}(WL)v_{OV}$ in C

(eq5.4) $$\frac{|Q|}{L} = C_{ox} W v_{OV} \text{ in } C/m$$

(eq5.5) $$|E| = \frac{v_{DS}}{L} \text{ in } V/m$$

(eq5.6) e-drift velocity $= \ldots$

$$\ldots = \mu_n |E| \text{ in } \frac{V}{m}\frac{m^2}{Vs} = \frac{m}{s}$$

5.1.4. Applying a Small v_{DS}

- **Q:** How does one calculate charge per unit length of *n*-

el.

Note that these two values may be employed to define current in amperes (aka. *C/s*).

- **A:** Note that v_{DS} establishes an electric field *E* across length of *n*-channel, this may calculate e-drift velocity.

action: divide both sides by L

(eq5.2) $|Q| = C_{ox}(WL)v_{OV}$ in C

(eq5.4) $\frac{|Q|}{L} = C_{ox}Wv_{OV}$ in C/m

(eq5.5) $|E| = \frac{v_{DS}}{L}$ in V/m

(eq5.6) e-drift velocity = …

$\ldots = \mu_n |E|$ in $\frac{V}{m}\frac{m^2}{Vs} = \frac{m}{s}$

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5.1.4. **Applying a Small v_{DS}**

- **Q:** What is observed from equation (5.7)?
  - **A:** For small values of v_{DS}, the *n*-channel acts like a variable resistance whose value is controlled by v_{OV}.

$$\text{(eq5.7)}\quad i_D = \left[(\mu_n C_{ox}) \frac{W}{L} v_{OV} \right] v_{DS} \text{ in } A$$

$$\text{(eq5.8a)}\quad r_{DS} = \frac{v_{DS}}{i_D} = \frac{1}{\underbrace{(\mu_n C_{ox})}_{\text{process transconductance parameter}} \left(\frac{W}{L} \right) v_{OV}} \text{ in } \Omega$$

aspect ratio

5.1.4. **Applying a Small v_{DS}**

Note that this v_{OV} represents the depth of the *n*-channel - what if it is not assumed to be constant? How does this equation change?

Note that this is one **VERY IMPORTANT** equation in Chapter 5.

n equation (5.7)?

v_{DS}, the *n*-channel acts like a

variable resistance whose value is controlled by v_{OV}.

$$\text{(eq5.7)}\quad i_D = \left[\left(\mu_n C_{ox}\right)\frac{W}{L}v_{OV}\right]v_{DS} \text{ in } A$$

$$\text{(eq5.8a)}\quad r_{DS} = \frac{v_{DS}}{i_D} = \frac{1}{\underbrace{\left(\mu_n C_{ox}\right)}_{\substack{\text{process}\\ \text{transconductance}\\ \text{parameter}}}\left(\frac{W}{L}\right)_{\substack{\text{aspect}\\ \text{ratio}}}v_{OV}} \text{ in } \Omega$$

5.1.4. **Applying a Small v_{DS}**

- **Q:** What three factors is r_{DS} dependent on?
  - **A: process transconductance parameter for NMOS** ($\mu_n C_{ox}$) – which is determined by the manufacturing process
  - **A: aspect ratio** (W/L) – which is dependent on size requirements / allocations
  - **A: overdrive voltage** (v_{OV}) – which is applied by the user

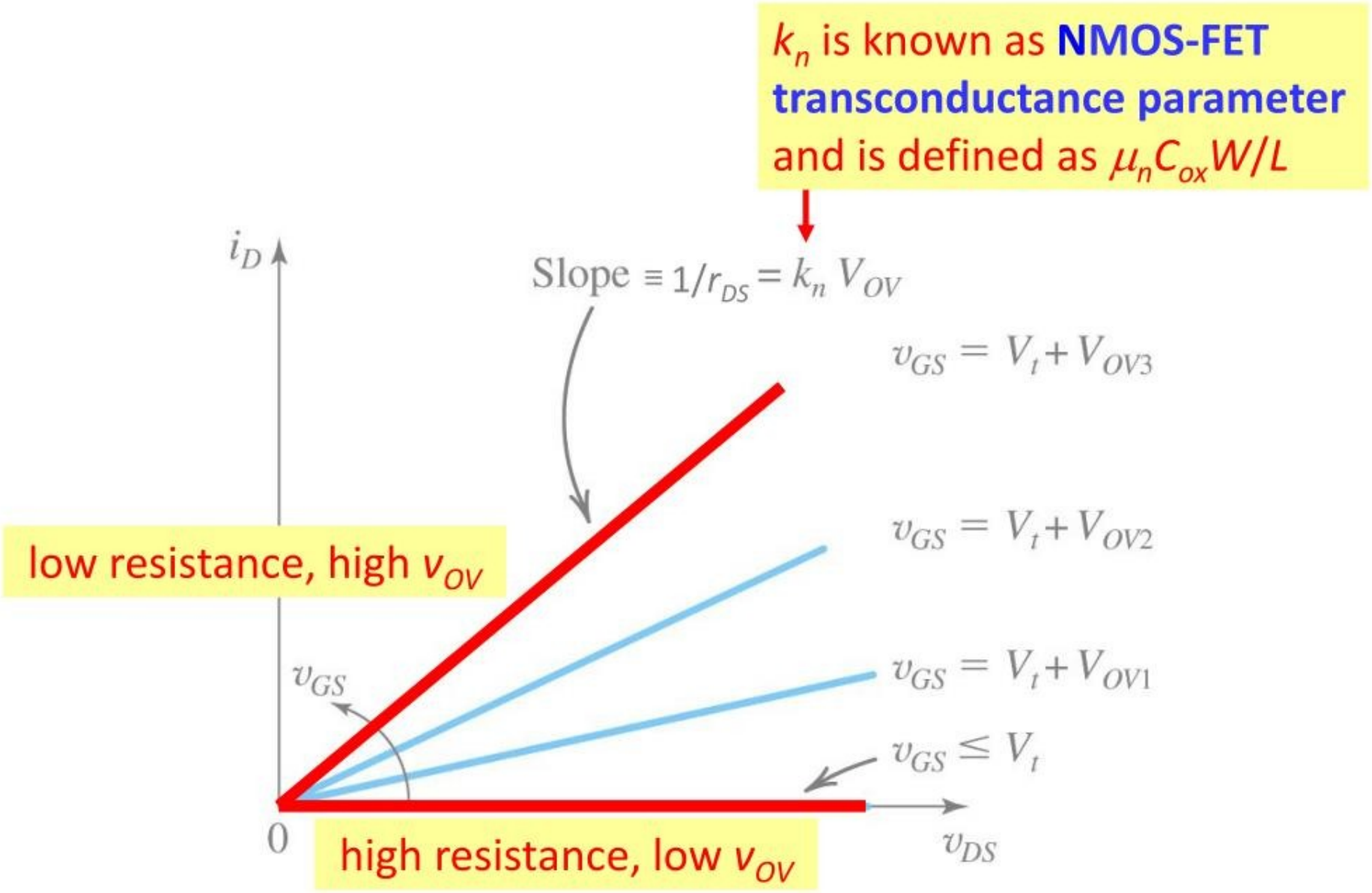


Figure 5.4: The i_D-v_{DS} characteristics of the MOSFET in Figure 5.3. when the voltage applied between drain and source V_{DS} is kept small.

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5.1.5. Operation as v_{DS} is Increased

- **Q:** What happens to i_D when v_{DS} increases beyond "small values"?
  - **A:** The relationship between them ceases to be linear.
- **Q:** How can this non-linearity be explained?
  - **step #1:** Assume that v_{GS} is held constant at value greater than V_t.
  - **step #2:** Also assume that v_{DS} is applied and appears as voltage drop across *n*-channel.
  - **step #3:** Note that voltage decreases from v_{GS} at the source end of channel to v_{GD} at drain end, where…
    - $v_{GD} = v_{GS} - v_{DS}$
    - $v_{GD} = V_t + v_{OV} - v_{DS}$

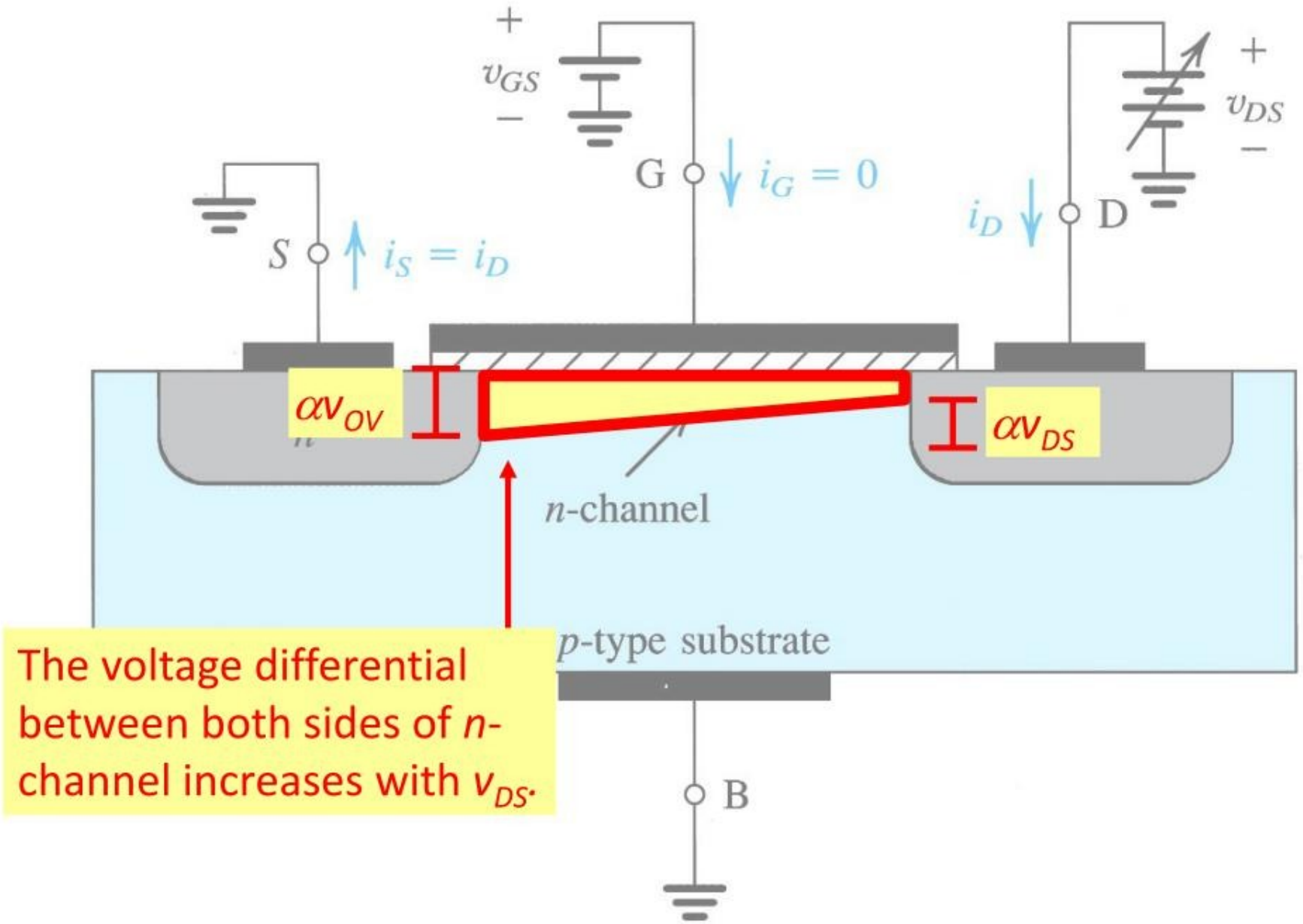


Figure 5.5: Operation of the e-NMOS transistor as v_{DS} is increased.

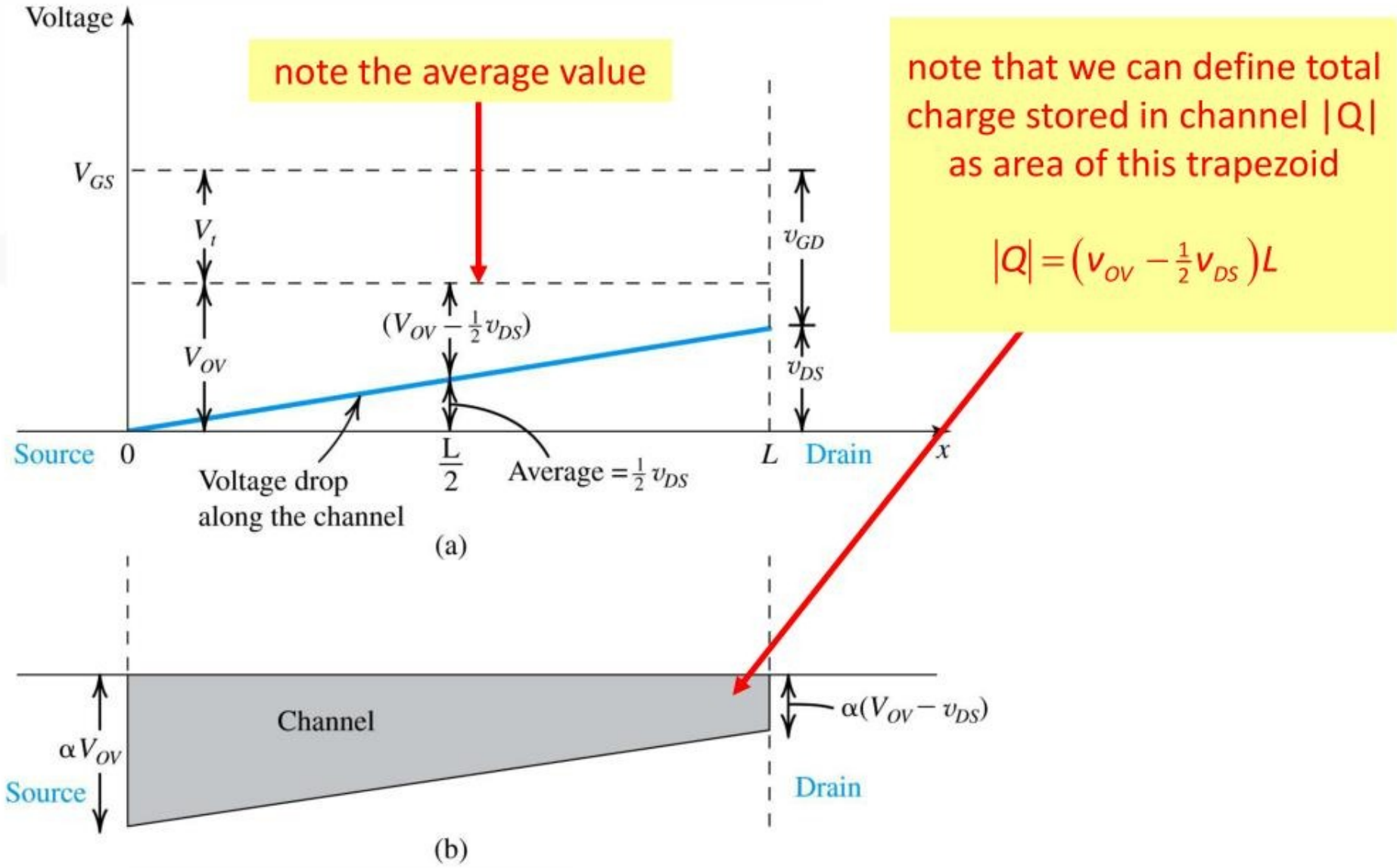


Figure 5.6(a): For a MOSFET with $v_{GS} = V_t + v_{OV}$ application of vDS causes the voltage drop along the channel to vary linearly, with an average value of vDS at the midpoint. Since $v_{GD} > V_t$, the channel still exists at the drain end. (b) The channel shape corresponding to the situation in (a). While the depth of the channel at the source is still proportional to v_{OV}, the drain end is not.

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Q: How can this non-linearity be explained?

- **step #4:** Define i_{DS} in terms of v_{DS} and v_{OV}.

(eq5.7) $$i_D = \left[(\mu_n C_{ox}) \frac{W}{L} \overbrace{\left(v_{OV} - \tfrac{1}{2} v_{DS}\right)}^{\text{action: replace } v_{OV} \text{ with } v_{OV} - \frac{1}{2} v_{DS}} \right] v_{DS}$$

$$i_D = \begin{cases} (\mu_n C_{ox}) \frac{W}{L} \left(v_{OV} - \tfrac{1}{2} v_{DS}\right) v_{DS} & \text{if } v_{DS} < v_{OV} \\ \underbrace{(\mu_n C_{ox}) \frac{W}{L} \left(v_{OV} - \tfrac{1}{2} v_{DS}\right) v_{DS}}_{\text{if } v_{DS} > v_{OV} \text{ then } v_{DS} = v_{OV}} & \text{otherwise} \end{cases}$$

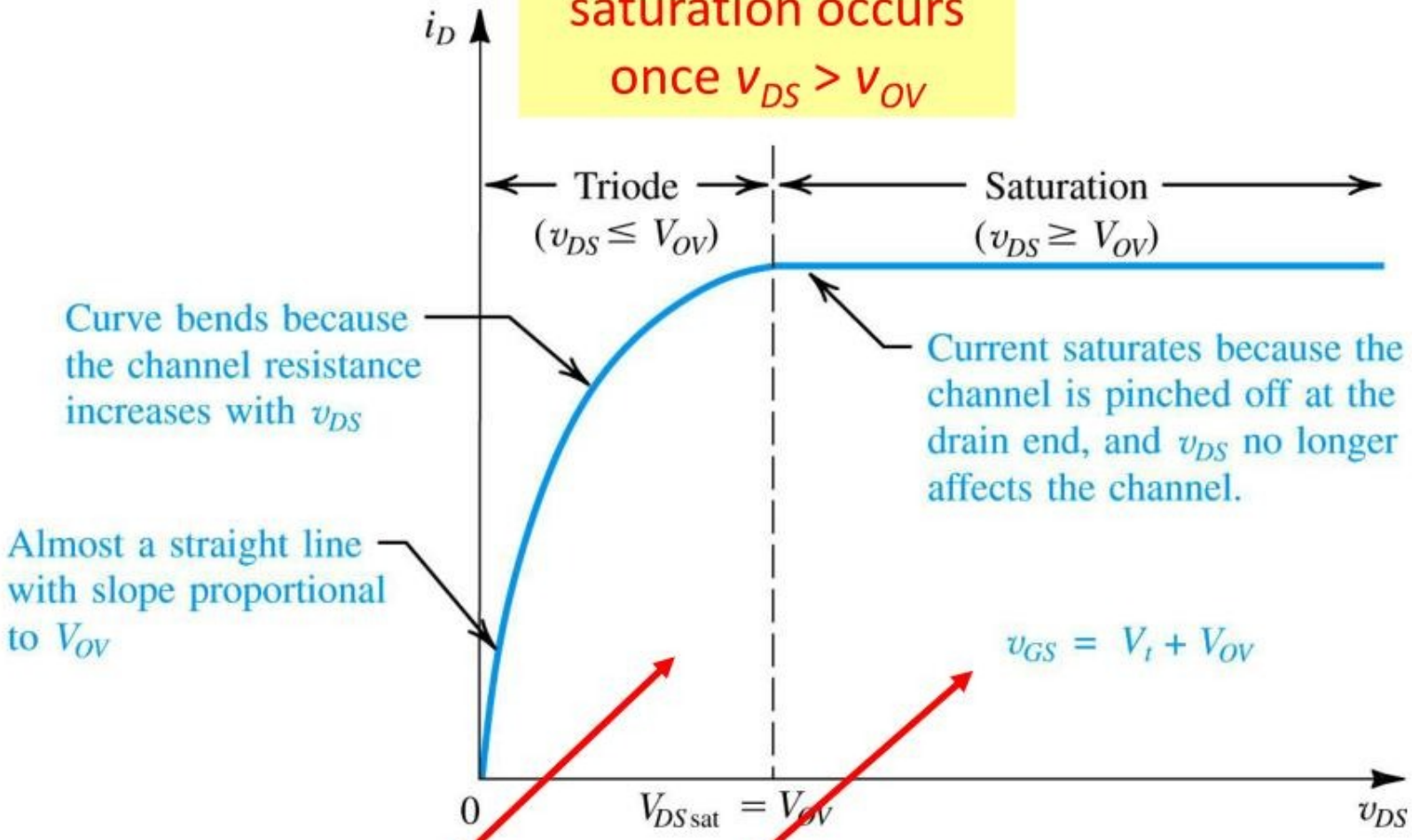
i_D is dependent on the apparent v_{OV} (not v_{DS} inherently) which does not change after $v_{DS} > v_{OV}$

(eq5.14) $$i_D = \begin{cases} (\mu_n C_{ox}) \frac{W}{L} \left(v_{OV} - \tfrac{1}{2} v_{DS}\right) v_{DS} & \text{if } v_{DS} < v_{OV} \\ \frac{1}{2} (\mu_n C_{ox}) \frac{W}{L} v_{OV}^2 & \text{otherwise} \end{cases}$$

in *A*

triode vs. saturation region

saturation occurs once $v_{DS} > v_{OV}$



(eq5.14) $i_D = \begin{cases} \text{triode: } (\mu_n C_{ox})\frac{W}{L}(v_{OV} - \frac{1}{2}v_{DS})v_{DS} & \text{if } v_{DS} < v_{OV} \\ \text{saturation: } \frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2 & \text{otherwise} \end{cases}$ in A

5.1.6. **Operation for** $v_{DS} >> v_{OV}$

- In section 5.1.5, we assume that *n*-channel is tapered but **channel pinch-off** does not occur.
  - Trapezoid doesn't become triangle for $v_{GD} > V_t$
- **Q:** What happens if $v_{DS} > v_{OV}$?
  - **A:** MOSFET enters **saturation region**. Any further increase in v_{DS} has no effect on i_D.

pinch-off does not mean blockage of current

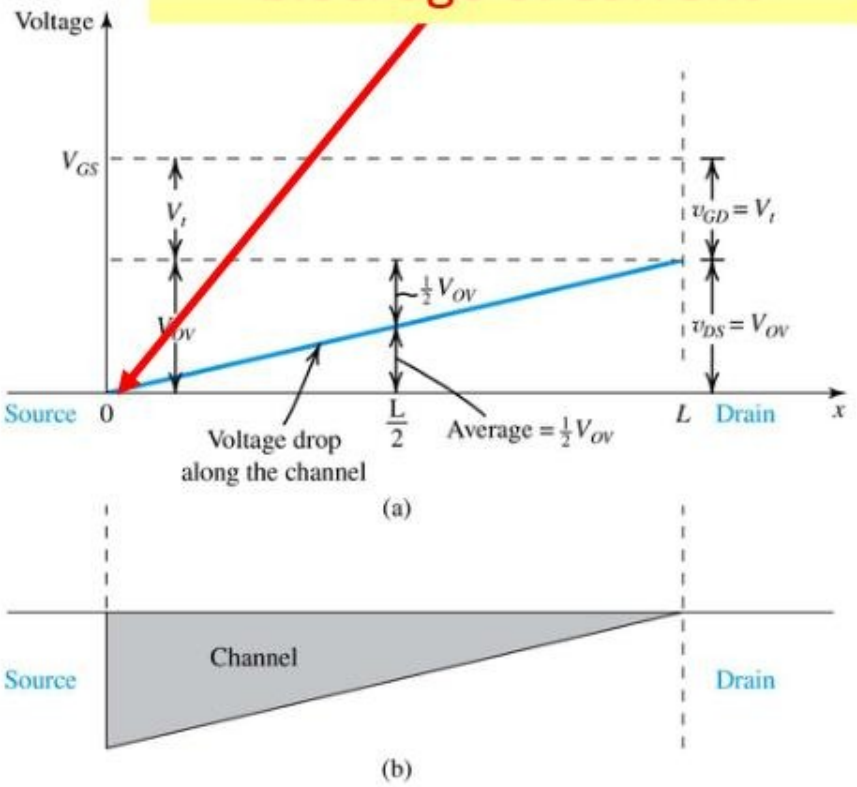
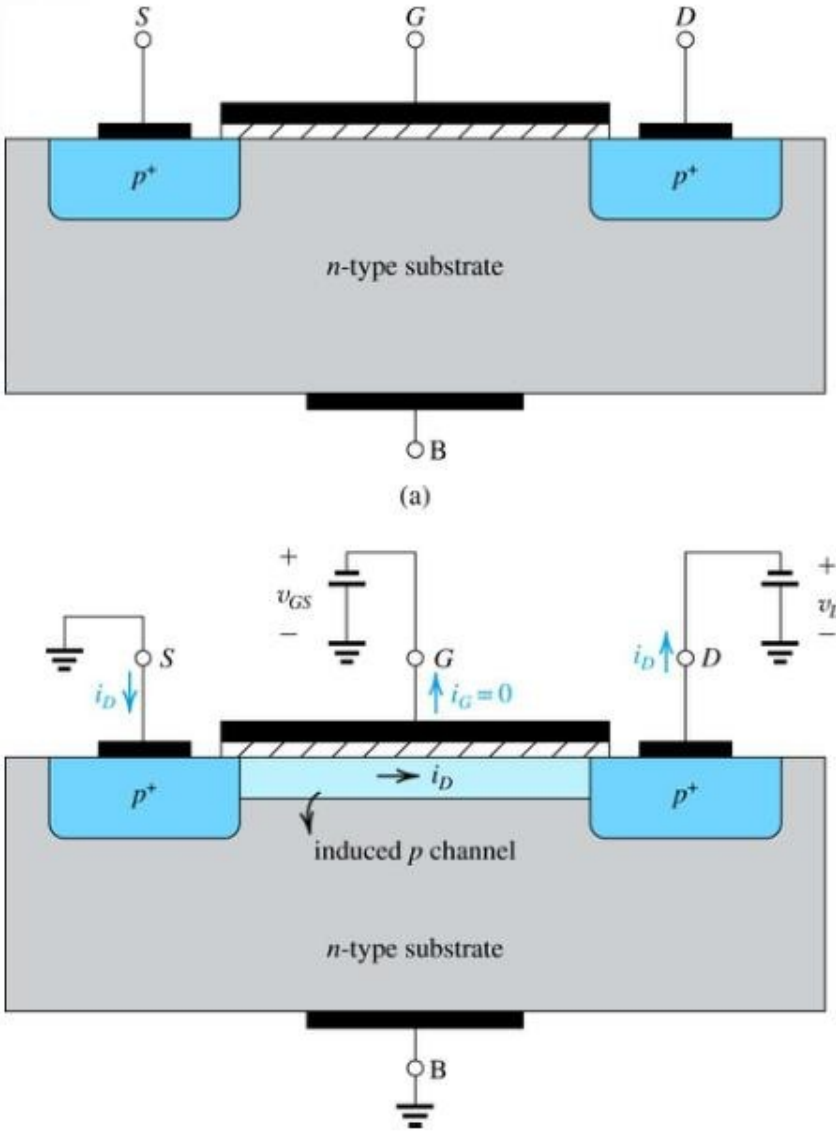


Figure 5.8: Operation of MOSFET with $v_{GS} = V_t + v_{OV}$ as v_{DS} is increased to v_{OV}. At the drain end, v_{GD} decreases to V_t and the channel depth at the drain-end reduces to zero (pinch-off). At this point, the MOSFET enters saturation more of operation. Further increasing v_{DS} (beyond v_{OV}) has no effect on the channel shape and i_D remains constant.

Example 5.1: NMOS MOSFET

- **Example 5.1. Problem Statement:** Consider an NMOS process technology for which $L_{min} = 0.4\mu m$, $t_{ox} = 8nm$, $\mu_n = 450 cm^2/Vs$, $V_t = 0.7V$.
- **Q(a):** Find C_{ox} and k'_n.
- **Q(b):** For a MOSFET with $W/L = 8\mu m/0.8\mu m$, calculate the values of v_{OV}, v_{GS}, and v_{DSmin} needed to operate the transistor in the saturation region with dc current $I_D = 100\mu A$.
- **Q(c):** For the device in (b), find the values of v_{OV} and v_{GS} required to cause the device to operate as a 1000*ohm* resistor for very small v_{DS}.

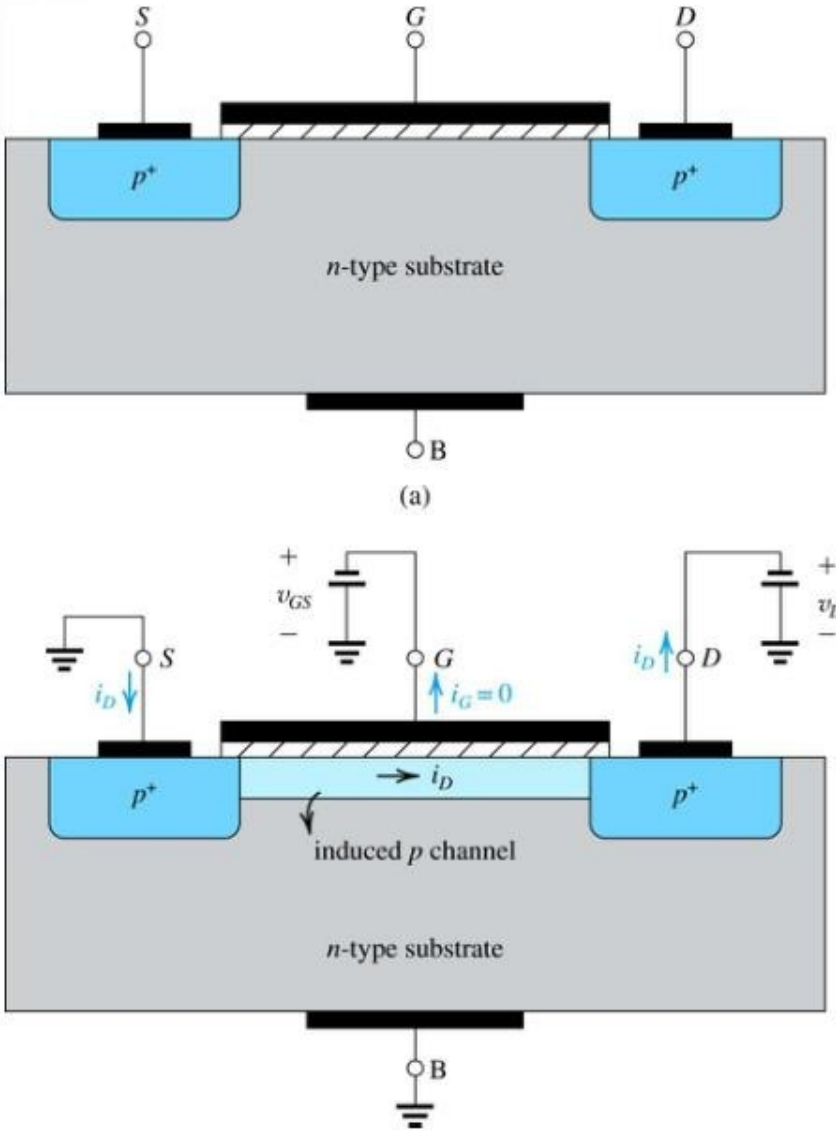
5.1.7. The *p*-Channel MOSFET



- Figure 5.9(a) shows cross-sectional view of a *p*-channel enhancement-type MOSFET.
  - structure is similar but "opposite" to *n*-channel
- **complementary devices** – two devices such as the *p*-channel and *n*-channel MOSFET's.

Figure 5.9(a): Physical structure of the PMOS transistor. Note that it is similar to the NMOS transistor shown in Figure 5.1(b), except that all semiconductor regions are reversed in polarity. **(b)** A negative voltage v_{GS} of magnitude greater than $|V_{tp}|$ induces a p-channel, and a negative v_{DS} causes a current i_D to flow from source to drain.

5.1.7. The *p*-Channel MOSFET



- **Q:** What are main differences between *n*-channel and *p*-channel?
  - **A:** Negative (not positive) voltage applied to gate "closes" the channel
    - allowing path for current flow
  - **A:** Threshold voltage (previously represented as V_t) is represented as V_{tp}
    - $|v_{GS}| > |V_{tp}|$ to close channel

Figure 5.9(a): Physical structure of the PMOS transistor. Note that it is similar to the NMOS transistor shown in Figure 5.1(b), except that all semiconductor regions are reversed in polarity. **(b)** A negative voltage v_{GS} of magnitude greater than $|V_{tp}|$ induces a p-channel, and a negative v_{DS} causes a current i_D to flow from source to drain.

5.1.7. The *p*-Channel MOSFET

- **Q:** What are main differences between *n*-channel and *p*-channel?
  - **A:** Process transconductance parameters are defined differently
    - $k'_p = \mu_p C_{ox}$
    - $k_p = \mu_p C_{ox}(W/L)$
  - **A:** The rest, essentially, is the same, but with reverse polarity...

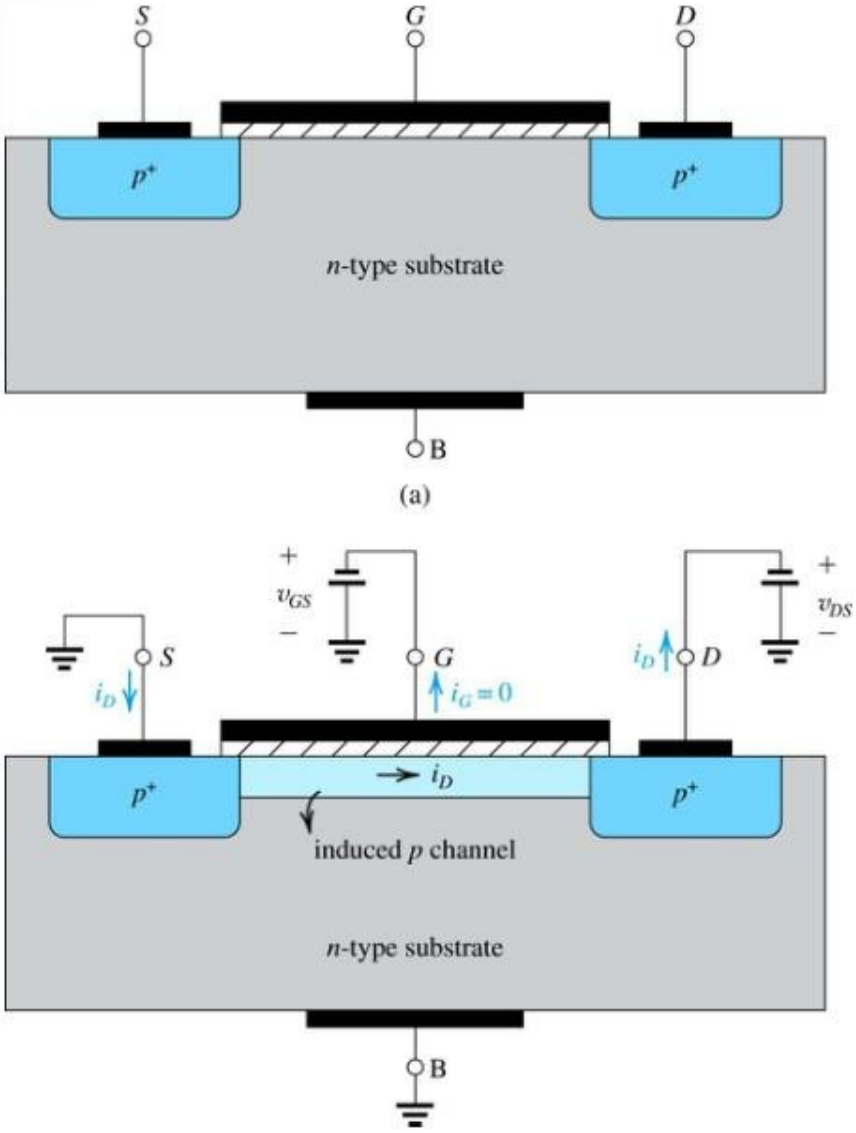


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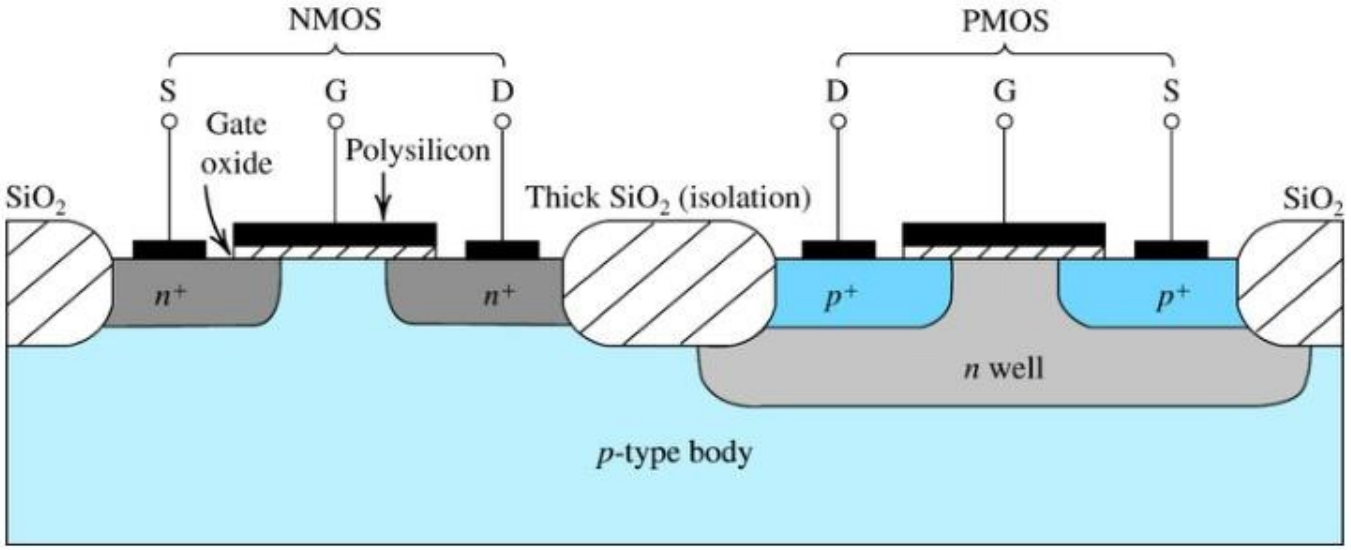
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5.1.7. The *p*-Channel MOSFET

- PMOS technology originally dominated the MOS field (over NMOS). However, as manufacturing difficulties associated with NMOS were solved, "they" took over
- **Q:** Why is NMOS advantageous over PMOS?
  - **A:** Because electron mobility μ_n is 2 – 4 times greater than hole mobility μ_p.
- **complementary MOS** (CMOS) **technology** – is technology which allows fabrication of both N and PMOS transistors on a single chip.

5.1.8. **Complementary MOS or CMOS**

- CMOS employs MOS transistors of both polarities.
  - more difficult to fabricate
  - more powerful and flexible
  - now more prevalent than NMOS or PMOS



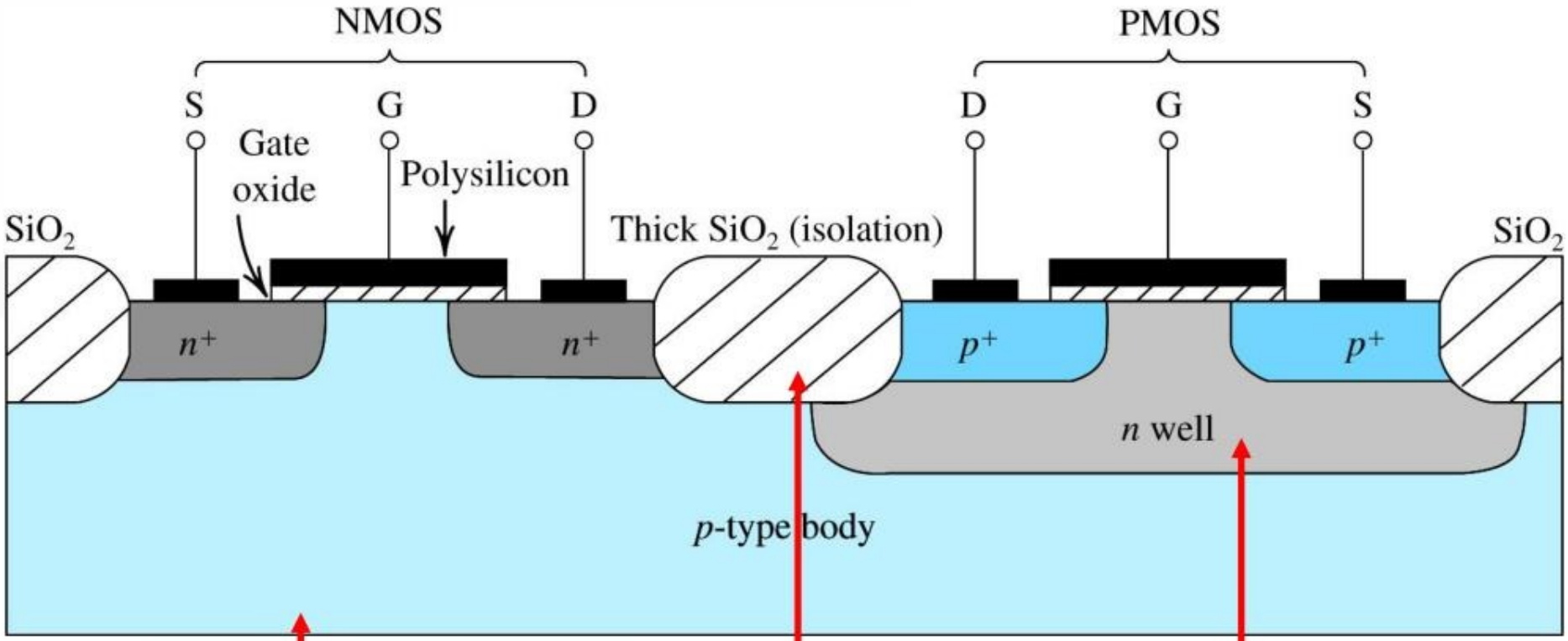


Figure 5.10: Cross-section of a CMOS integrated circuit. Note that the PMOS transistor is formed in a separate *n*-type region, known as an *n* well. Another arrangement is also possible in which an *n*-type body is used and the *n* device is formed in a *p* well. Not shown are the connections made to the *p*-type body and to the *n* well; the latter functions as the body terminal for the *p*-channel device.

p-type semiconductor provides the MOS body (and allows generation of n-channel)

SiO_2 is used to isolate NMOS from PMOS

n-well is added to allow generation of *p*-channel

Quick Recap!

- The equation used to define i_D depends on relationship btw v_{DS} and v_{OV}.
  - $v_{DS} << v_{OV}$
  - $v_{DS} < v_{OV}$
  - $v_{DS} => v_{OV}$
  - $v_{DS} >> v_{OV}$

μ_n represents mobility of electrons at surface of the n-channel in m^2 / Vs

(eq5.7) $i_D = \underbrace{(C_{ox} W v_{OV})}_{\text{charge per unit length of } n\text{-channel in } C/m} \underbrace{\left(\frac{\mu_n v_{DS}}{L}\right)}_{\text{electron drift velocity in } m^2/Vs}$ in A

(eq5.14) $i_D = (\mu_n C_{ox}) \frac{W}{L} (v_{OV} - \frac{1}{2} v_{DS}) v_{DS}$ in A

(eq5.17) $i_D = \frac{1}{2} (\mu_n C_{ox}) \frac{W}{L} v_{OV}^2$ in A

(e This has not been covered yet! A

5.2. Current-Voltage Characteristics

- Figure 5.11. shows an *n*-channel enhancement MOSFET.
- There are four terminals:
  - drain (D), gate (G), body (B), and source (S).
- Although, it is assumed that body and source are connected.

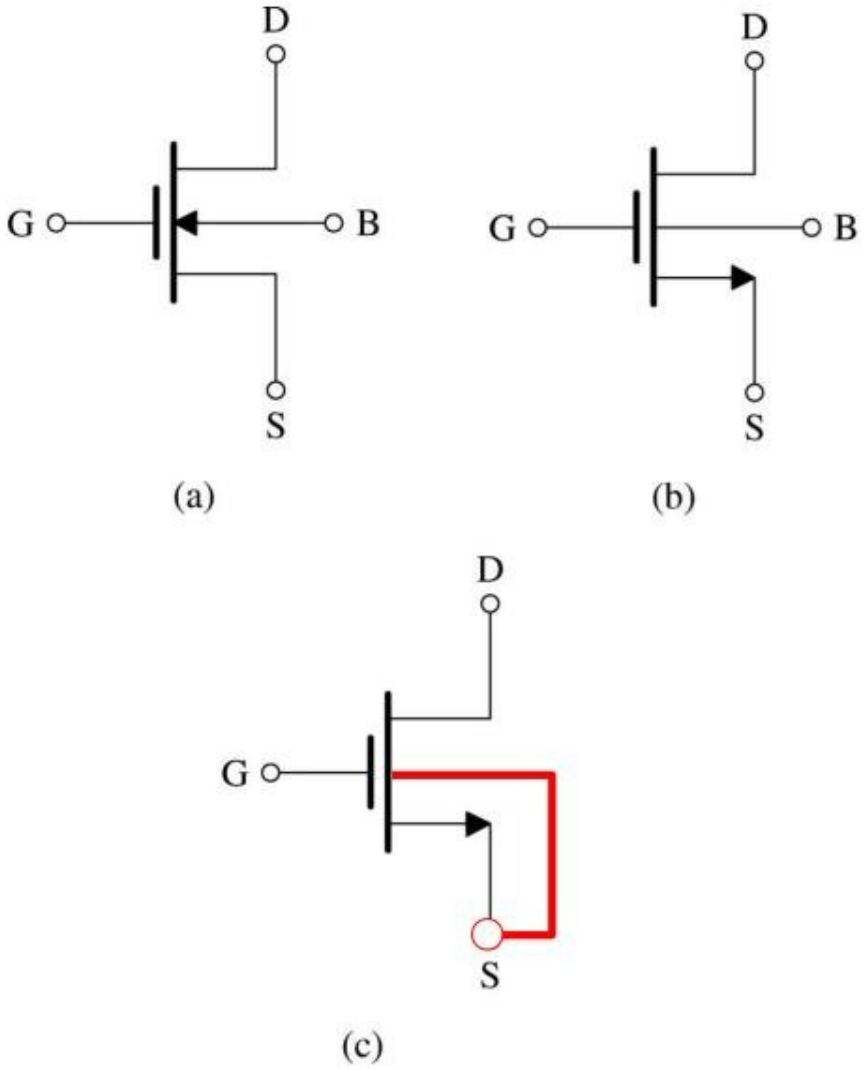
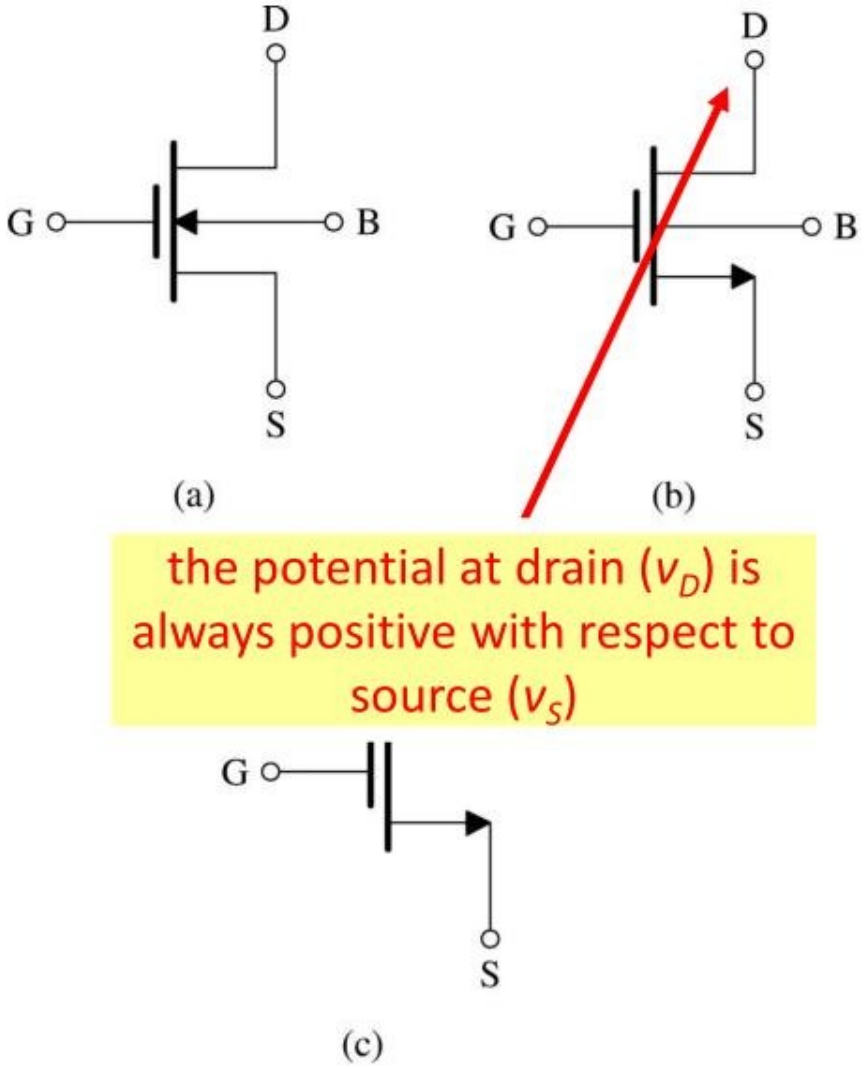


Figure 5.11 (a): Circuit symbol for the *n*-channel enhancement-type MOSFET. **(b)** Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., *n* channel). **(c)** Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

5.2. Current-Voltage Characteristics

- Although MOSFET is symmetrical device, one often designates terminals as source and drain.
- **Q:** How does one make this designation?
  - **A:** By polarity of voltage applied.
- Arrowheads designate "normal" direction of current flow
  - Note that, in part (b), we designate current as D→S.
  - No need to place arrow with B.



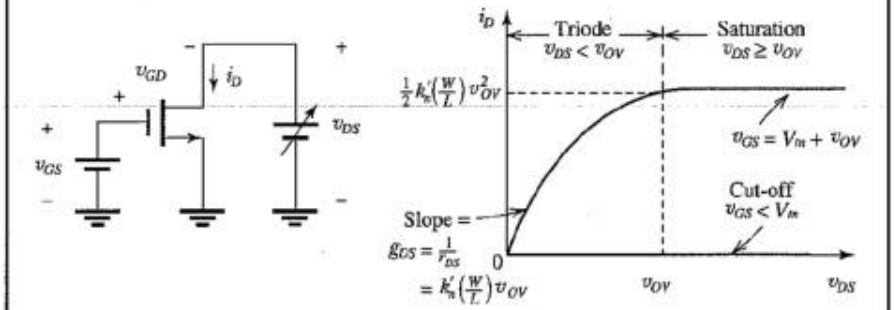
the potential at drain (v_D) is always positive with respect to source (v_S)

Figure 5.11 (a): Circuit symbol for the *n*-channel enhancement-type MOSFET. **(b)** Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., *n* channel). **(c)** Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

5.2.2. The i_D-v_{DS} Characteristics

- Table 5.1. provides a compilation of the conditions and formulas for operation of NMOS transistor in three regions.
  - cutoff
  - triode
  - saturation

Table 5.1 Regions of Operation of the Enhancement NMOS Transistor



- $v_{GS} < V_{tn}$: no channel; transistor in cut-off; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$: a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

Triode Region

Continuous channel, obtained by:

$v_{GS} \geq V_{tn}$

or equivalently:

$v_{DS} < v_{OV}$

Then,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[(v_{GS} - V_{tn})v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

or equivalently,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right)v_{DS}$$

Saturation Region

Pinched-off channel, obtained by:

$v_{GD} \leq V_{tn}$

or equivalently:

$v_{DS} \geq v_{OV}$

Then

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2$$

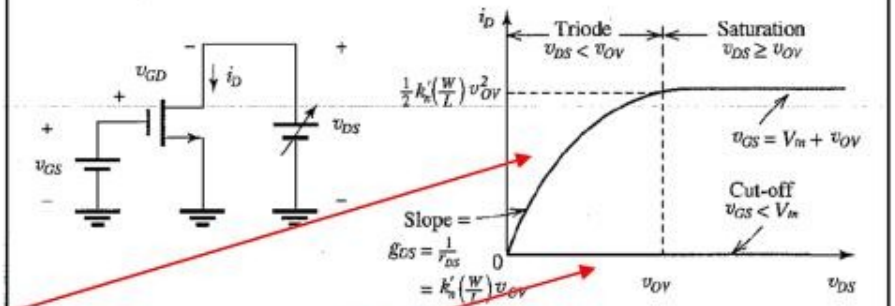
or equivalently,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

5.2.2. The i_D-v_{DS} Characteristics

- At top of table, it shows circuit consisting of NMOS transistor and two dc supplies (v_{DS}, v_{GS})
- This circuit is used to demonstrate i_D-v_{DS} characteristic
  - 1st set v_{GS} to desired constant
  - 2nd vary v_{DS}
- Two curves are shown...
  - $v_{GS} < V_{tn}$
  - $v_{GS} = V_{tn} + v_{OV}$

Table 5.1 Regions of Operation of the Enhancement NMOS Transistor



- $v_{GS} < V_{tn}$: no channel; transistor in cut-off; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$: a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

Triode Region

Continuous channel, obtained by:

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or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[(v_{GS} - V_{tn})v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

or equivalently,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right)v_{DS}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{tn}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

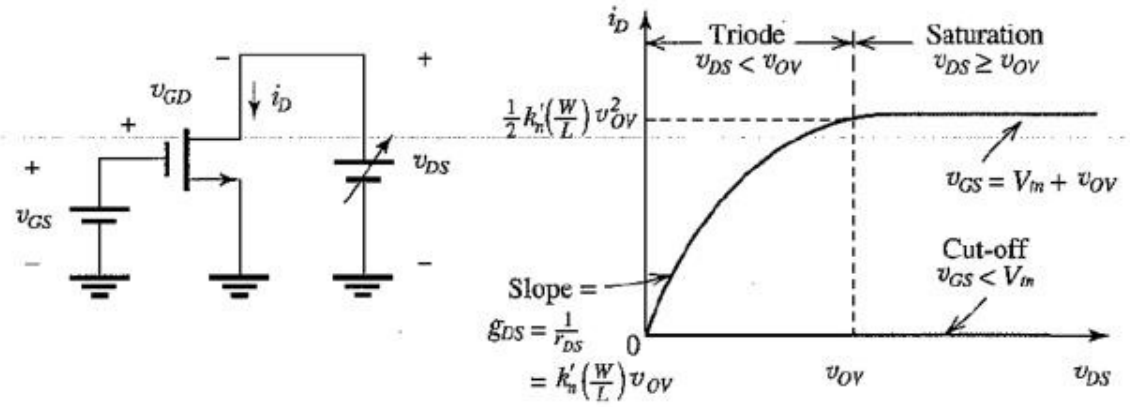
Then

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2$$

or equivalently,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

Table 5.1 Regions of Operation of the Enhancement NMOS Transistor



- $v_{GS} < V_{tn}$: no channel; transistor in cut-off; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$: a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

Triode Region

Continuous channel, obtained by:

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or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[(v_{GS} - V_{tn})v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

or equivalently,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right)v_{DS}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{tn}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

Then

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2$$

or equivalently,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

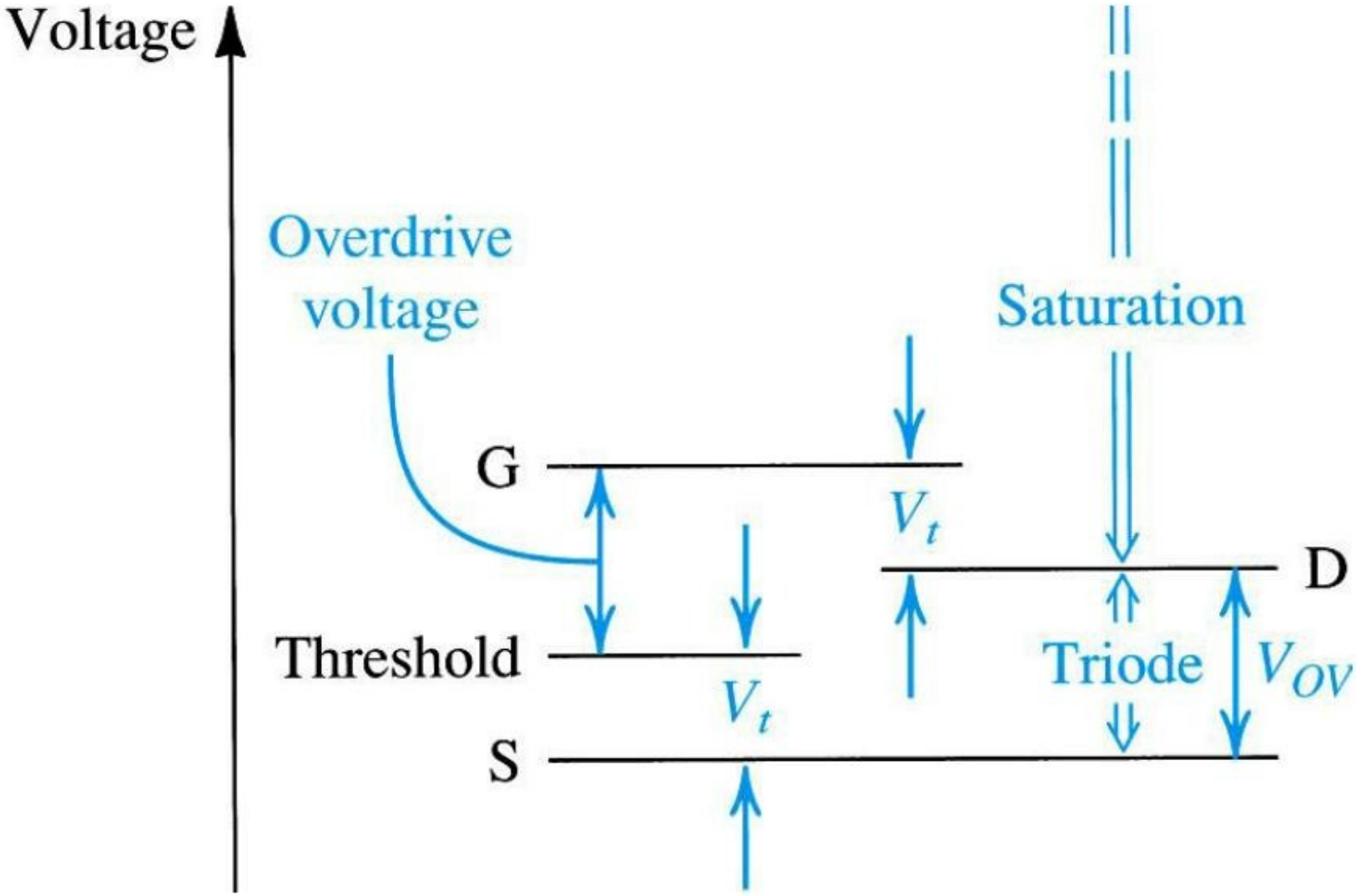


Figure 5.12: The relative levels of the terminal voltages of the enhancement NMOS transistor for operation in the triode region and in the saturation region.

equation (5.14)

as v_{GS} increases, so do the **(1)** saturation current and **(2)** beginning of the saturation region

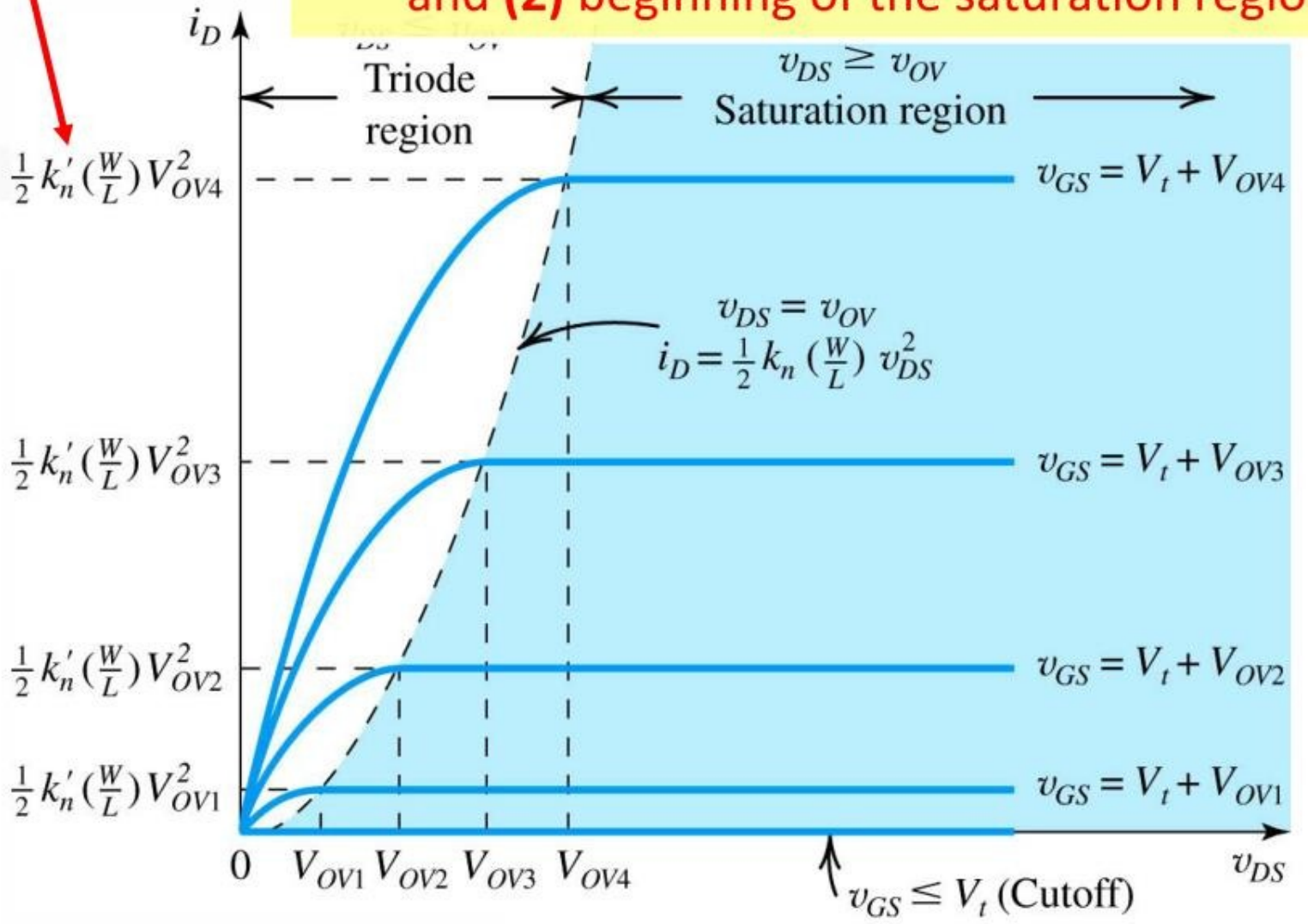


Figure 5.13: The $i_D - v_{DS}$ characteristics for an enhancement-type NMOS transistor

5.2.2. The i_D-v_{GS} Characteristic

- **Q:** When MOSFET's are employed to design amplifier, in what range will they be operated?
  - **A:** saturation
- In saturation, the drain current (i_D) is…
  - dependent on v_{GS}
  - independent of v_{DS}
- In effect, it becomes a voltage-controlled current source.
  - This is key for amplification.

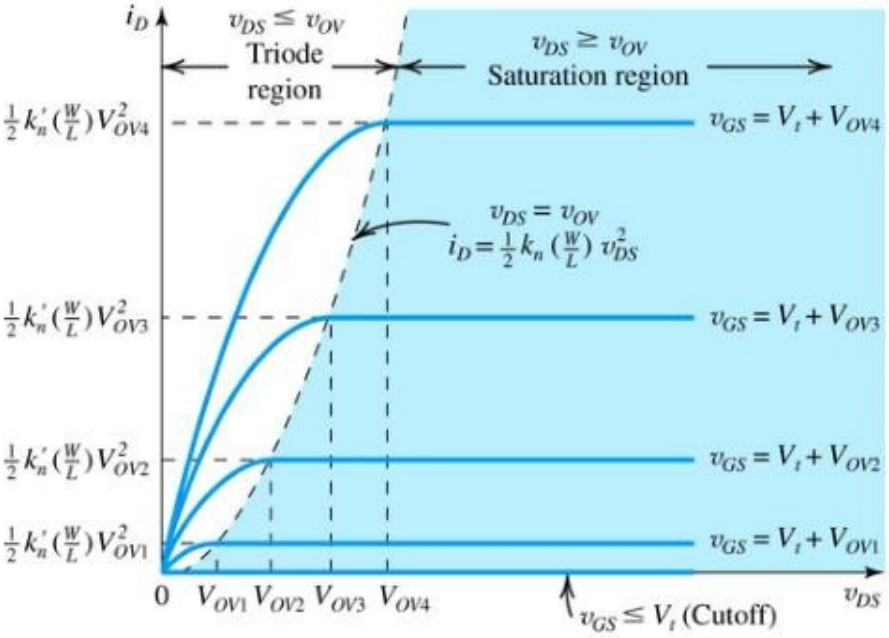


Figure 5.13: The $i_D - v_{DS}$ characteristics for an enhancement-type NMOS transistor

5.2.2. The i_D-v_{GS} Characteristic

- **Q:** What is one problem with (5.21)?
  - **A:** It is nonlinear w/ respect to v_{OV} … however, this is not of concern now.

- In effect, it becomes a voltage-controlled current source.
  - This is key for amplification.
  - Refer to (5.21).

(eq5.21) $$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\overbrace{(v_{GS}-V_{tn})^2}^{v_{OV}^2}$$

this relationship provides basis for application of MOSFET as amplifier

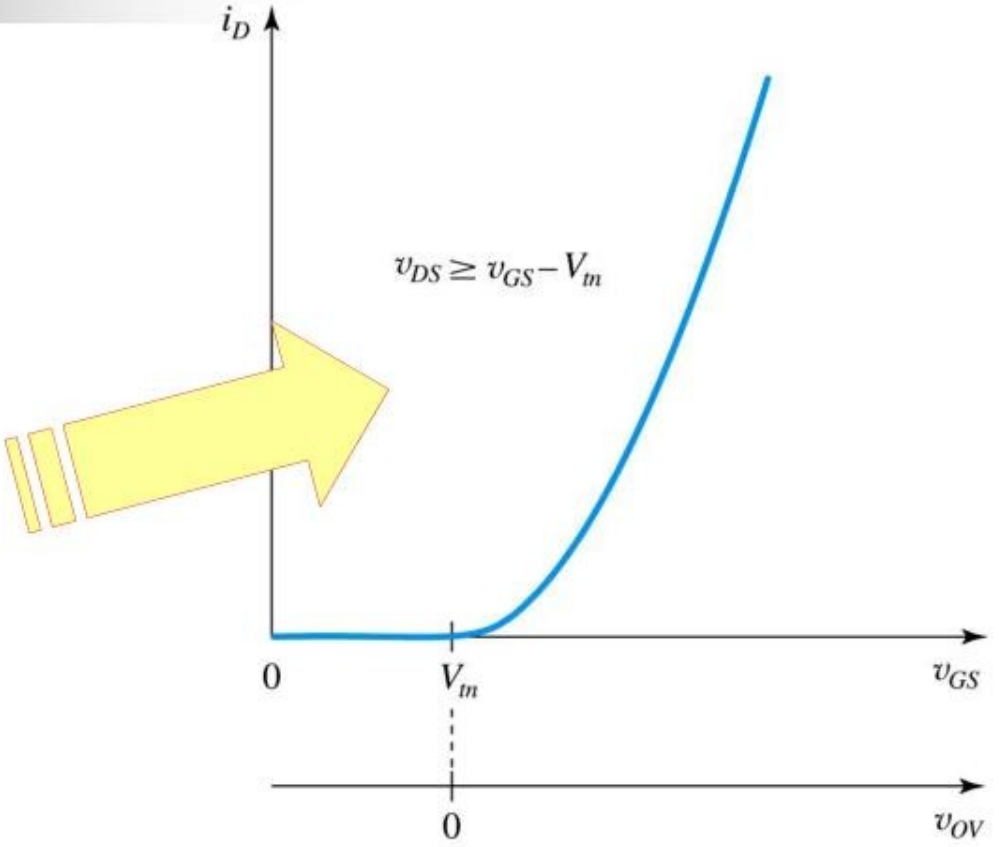


Figure 5.14: The i_D-v_{GS} characteristic of an NMOS transistor operating in the saturation region. The i_D-v_{OV} characteristic can be obtained by simply re-labeling the horizontal axis, that is, shifting the origin to the point $v_{GS} = V_{tn}$.

5.2.2. The i_D-v_{GS} Characteristic

- The view of transistor as CVCS is exemplified in figure 5.15.
  - This circuit is known as the **large-signal equivalent circuit.**
  - Current source is ideal.
  - Infinite output resistance represents independent, in saturation, of i_D from v_{DS}.

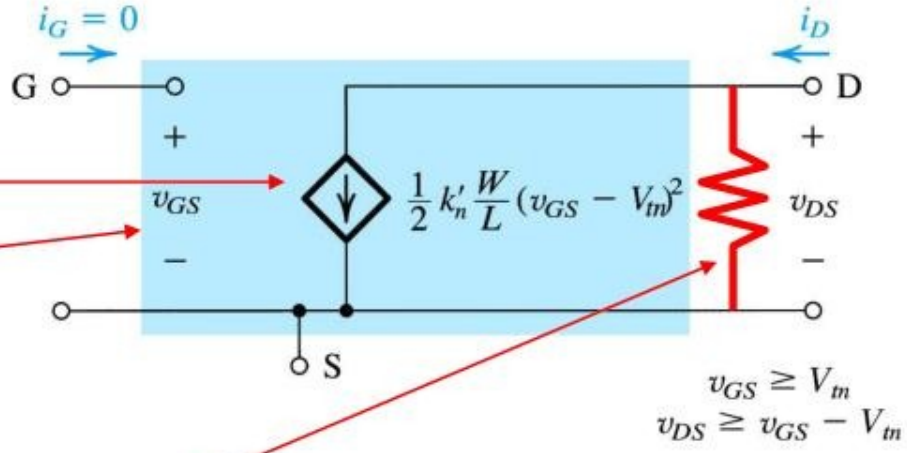


Figure 5.15: Large-signal equivalent-circuit model of an *n*-channel MOSFET operating in the saturation

note that, in this circuit, i_D is completely independent of v_{DS} (because no shunt resistor exists)

Example 5.2: NMOS Transistor

- **Example 5.2. Problem Statement:** Consider an NMOS transistor fabricated in an 0.18-μm process with L = 0.18μm and W = 2μm. The process technology is specified to have C_{ox} = 8.6$fF/\mu m^2$, μ_n = 450cm^2/Vs, and V_{tn} = 0.5V.
- **Q(a):** Find V_{GS} and V_{DS} that result in the MOSFET operating at the edge of saturation with ID = 100μA.
- **Q(b):** If VGS is kept constant, find VDS that results in I_D = 50μA.
- **Q(c):** To investigate the use of the MOSFET as a linear amplifier, let it be operating in saturation with V_{DS} = 0.3V. Find the change in i_D resulting from v_{GS} changing from 0.7V by +0.01V and -0.01V.

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5.2.4. Finite Output Resistance in Saturation

- In previous section, we assume (in saturation) i_D is independent of v_{DS}.
- Therefore, a change Δv_{DS} causes no change in i_D.
  - This implies that the incremental resistance R_S is infinite.
  - It is based on the idealization that, once the *n*-channel is pinched off, changes in v_{DS} will have no effect on i_D.
  - The problem is that, in practice, this is not completely true.

5.2.4. **Finite Output Resistance in Saturation**

- **Q:** What effect will increased v_{DS} have on *n*-channel once pinch-off has occurred?
  - **A:** It will cause the pinch-off point to move slightly away from the drain & create new depletion region.
  - **A:** Voltage across the (now shorter) channel will remain at (v_{OV}).
  - **A:** However, the additional voltage applied at v_{DS} will be seen across the "new" depletion region.

5.2.4. **Finite Output Resistance in Saturation**

this is the most important point here

- **Q:** What effect will increased v_{DS} have on *n*-channel once pinch-off has occurred?
  - **A:** This voltage accelerates electrons as they reach the drain end, and sweep them across the "new" depletion region.
  - **A:** However, at the same time, the length of the *n*-channel will decrease.
    - Known as **channel length modulation**.

5.2.4. Finite Output Resistance in Saturation

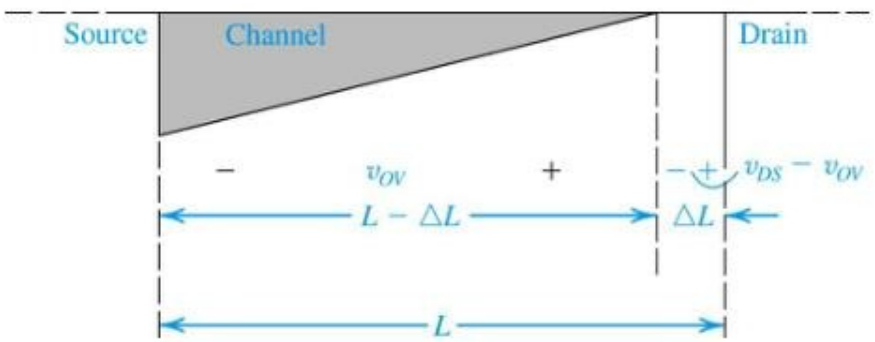


Figure 5.16: Increasing v_{DS} beyond v_{DSsat} causes the channel pinch-off point to move slightly away from the drain, thus reducing the effective channel length by ΔL

- **Q:** How do we account for "this effect" in i_D?
  - **A:** Refer to (5.23).

valid when $v_{DS} > v_{OV}$

(eq5.17) $$i_D = \frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2 \text{ in } A$$

(eq5.23) $$i_D = \frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2(1+\lambda v_{DS}) \text{ in } A$$

valid when $v_{DS} >> v_{OV}$

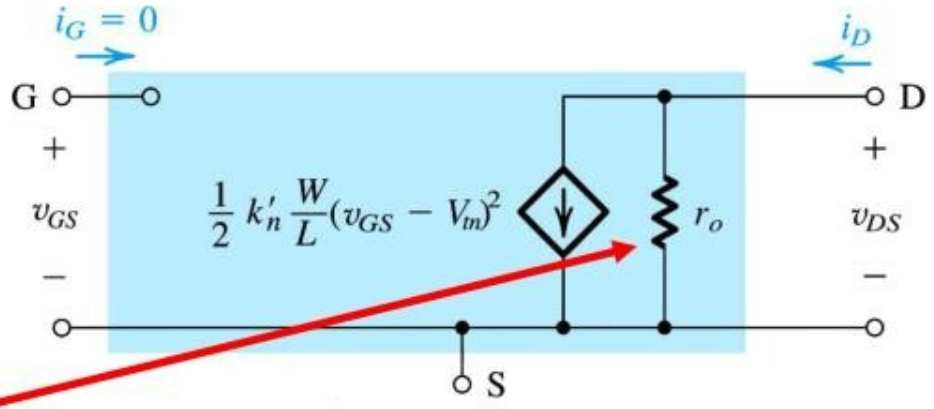


Figure 5.18: Large-Signal Equivalent Model of the *n*-channel MOSFET in saturation, incorporating the output resistance r_o. The output resistance models the linear dependence of i_D on v_{DS} and is given by (5.23)

- **A:** Addition of finite output resistance (r_o).

5.2.4. Finite Output Resistance in Saturation

- **Q:** How is r_o defined?
  - **step #1:** Note that r_o is the 1/slope of i_D-v_{DS} characteristic.
  - **step #2:** Define relationship between i_D and v_{DS} using (5.23).
  - **step #3:** Take derivative of this function.
  - **step #4:** Use above to define r_o.
- Note that r_o may be defined in terms of i_D, where i_D does not take in to account channel length modulation…

(eq5.24) $$r_o = \left[\frac{\partial i_D}{\partial v_{DS}}\right]^{-1}_{v_{GS}=constant}$$

$$\frac{\partial i_D}{\partial v_{DS}} = \frac{\partial}{\partial v_{DS}}\left(\overbrace{\frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2(1+\lambda v_{DS})}^{(5.23)}\right)$$

$$= \frac{\partial}{\partial v_{DS}}\left(\overbrace{\frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2(1+\lambda v_{DS})}^{(5.23)}\right)$$

$$= \frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2\lambda$$

(eq5.25) $$r_o = \left[\frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2\lambda\right]^{-1}_{v_{GS}=constant}$$

$$= \frac{1}{\lambda i_D} = \frac{V_A}{i_D}$$

5.2.4. Finite Output Resistance in Saturation

- **Q:** What is λ?
  - **A:** A device parameter with the units of V^{-1}, the value of which depends on manufacturer's design and manufacturing process.
    - much larger for newer tech's
- Figure 5.17 demonstrates the effect of channel length modulation on v_{DS}-i_D curves
  - In short, we can draw a straight line between V_A and saturation.

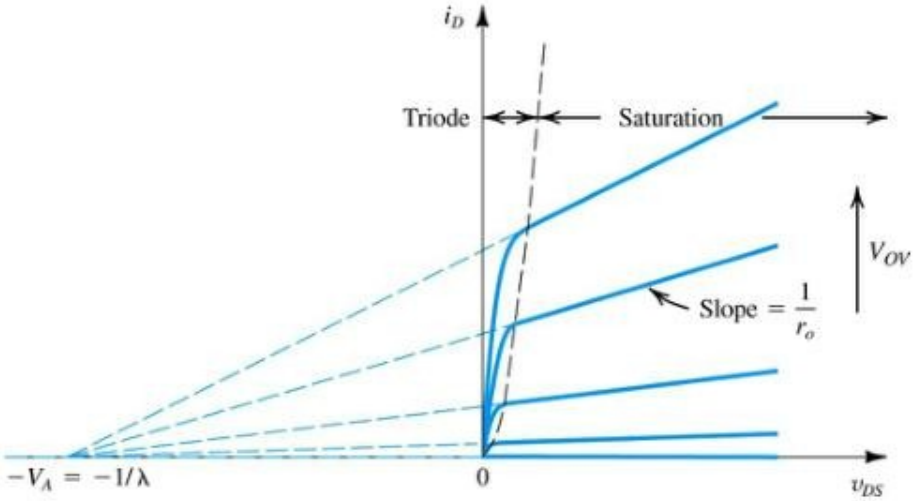
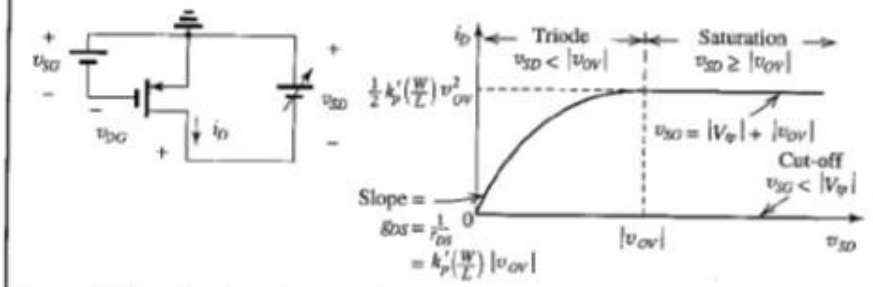


Figure 5.17: Effect of v_{DS} on i_D in the saturation region. The MOSFET parameter V_A depends on the process technology and, for a given process, is proportional to the channel length L.

5.2.5. **Characteristics of the *p*-channel MOSFET**

- Characteristics of the *p*-channel MOSFET are similar to the *n*-channel, however with many signs reversed.
- Please review section 5.2.5 from the text, with focus on table 5.2.

Table 5.2 Regions of Operation of the Enhancement PMOS Transistor

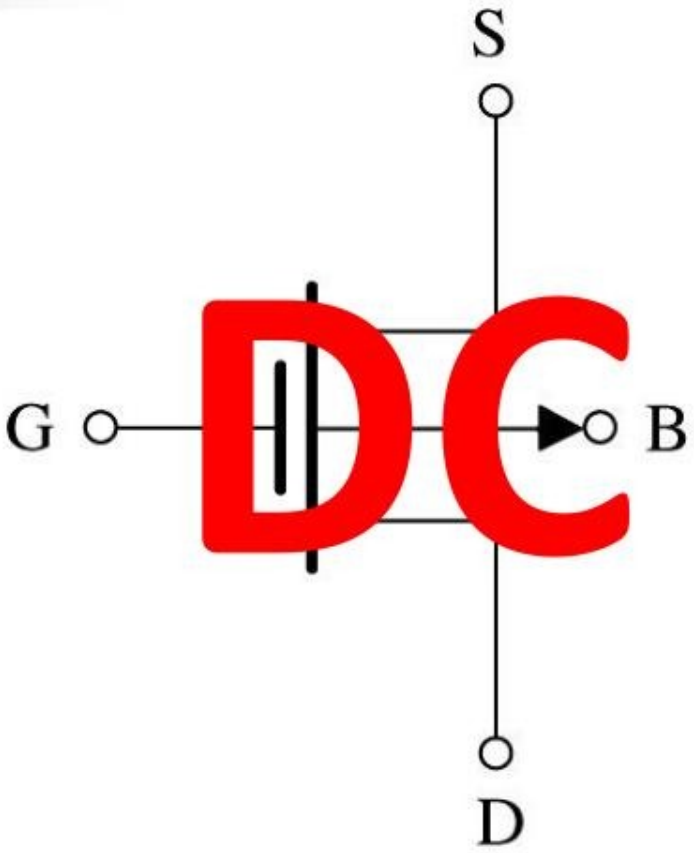


- $v_{SG} < |V_{tp}|$: no channel; transistor is cut-off; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$: a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

Triode Region	*Saturation Region*
Continuous channel, obtained by: $v_{DG} > \|V_{tp}\|$	Pinched-off channel, obtained by: $v_{DG} \leq \|V_{tp}\|$
or equivalently: $v_{SD} < \|v_{OV}\|$	or equivalently $v_{SD} \geq \|v_{OV}\|$
Then, $i_D = k_p'\left(\frac{W}{L}\right)\left[(v_{SG} - \|V_{tp}\|)v_{SD} - \frac{1}{2}v_{SD}^2\right]$	Then $i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(v_{SG} - \|V_{tp}\|)^2$
or equivalently $i_D = k_p'\left(\frac{W}{L}\right)\left(\|v_{OV}\| - \frac{1}{2}v_{SD}\right)v_{SD}$	or equivalently $i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)v_{OV}^2$

5.3. **MOSFET Circuits at DC**

- We move on to discuss how MOSFET's behave in dc circuits.
- We will neglect the effects of channel length modulation (assuming $\lambda = 0$).
- We will work in terms of overdrive voltage (v_{OV}), which reduces need to distinguish between PMOS and NMOS.



Example 5.3: NMOS Transistor

- **Problem Statement:** Design the circuit of Figure 5.21, that is, determine the values of R_D and R_S – so that the transistor operates at $I_D = 0.4mA$ and $V_D = +0.5V$. The NMOS transistor has $V_t = 0.7V$, $\mu_n C_{ox} = 100\mu A/V^2$, L = $1\mu m$, and W = $32\mu m$. Neglect the channel-length modulation effect (i. e. assume that $\lambda = 0$).

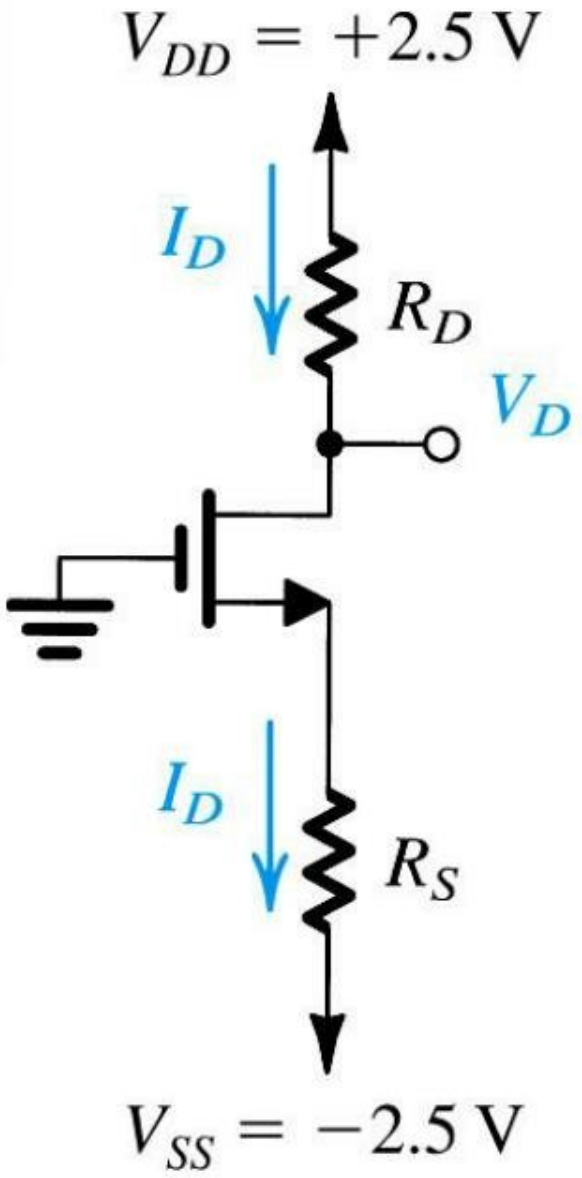
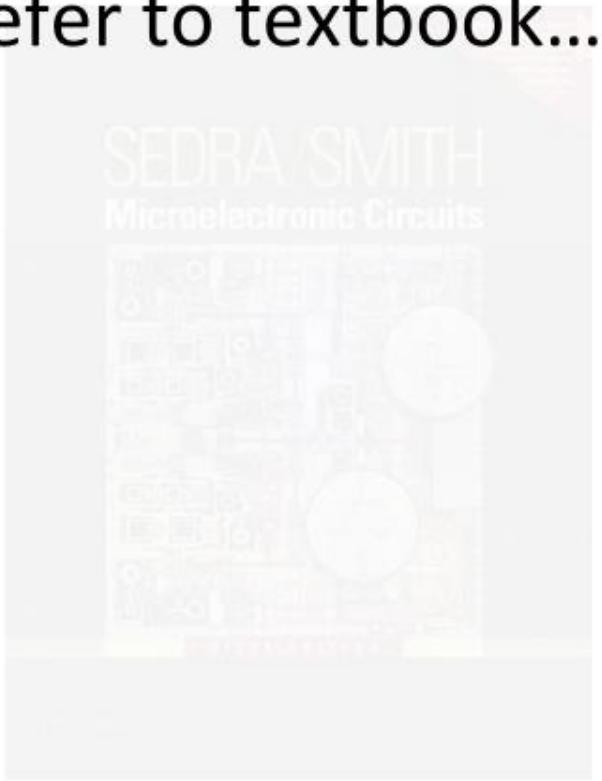


Figure 5.21: Circuit for Example 5.3.

Example 5.4:

- Refer to textbook…



Example 5.5: MOSFET

- **Problem Statement:** Design the circuit in Figure 5.23 to establish a drain voltage of 0.1*V*. What is the effective resistance between drain and source at this operating point? Let $V_{tn} = 1V$ and $k'_n(W/L) = 1mA/V^2$.

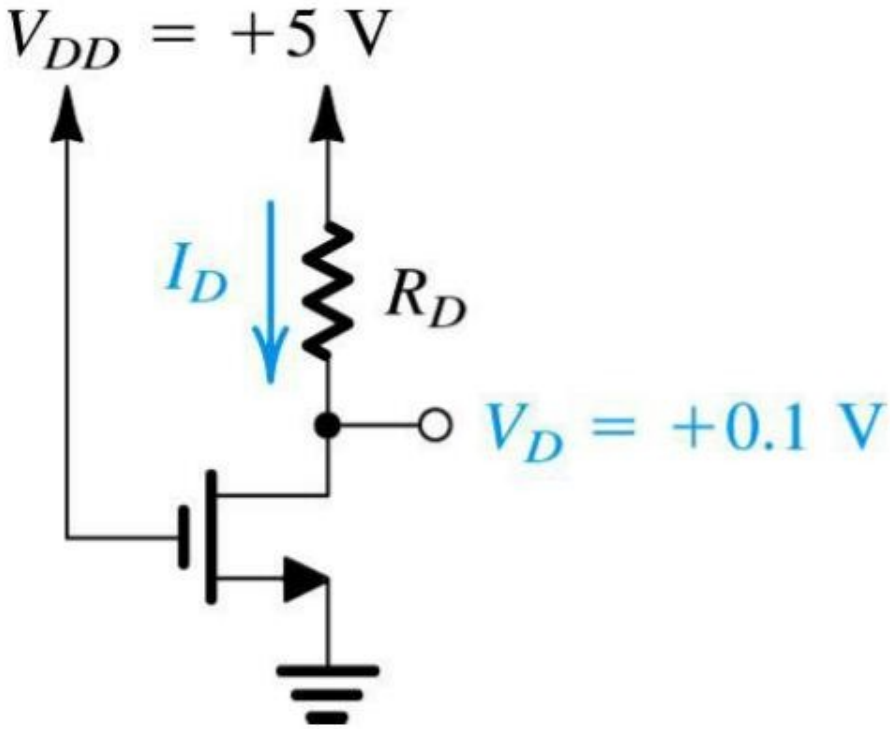


Figure 5.23: Circuit for Example 5.5.

Example 5.6: MOSFET

- **Problem Statement:** Analyze the circuit shown in Figure 5.24(a) to determine the voltages at all nodes and the current through all branches. Let $V_{tn} = 1V$ and $k'_n(W/L) = 1mA/V^2$. Neglect the channel-length modulation effect (i.e. assume $\lambda = 0$).

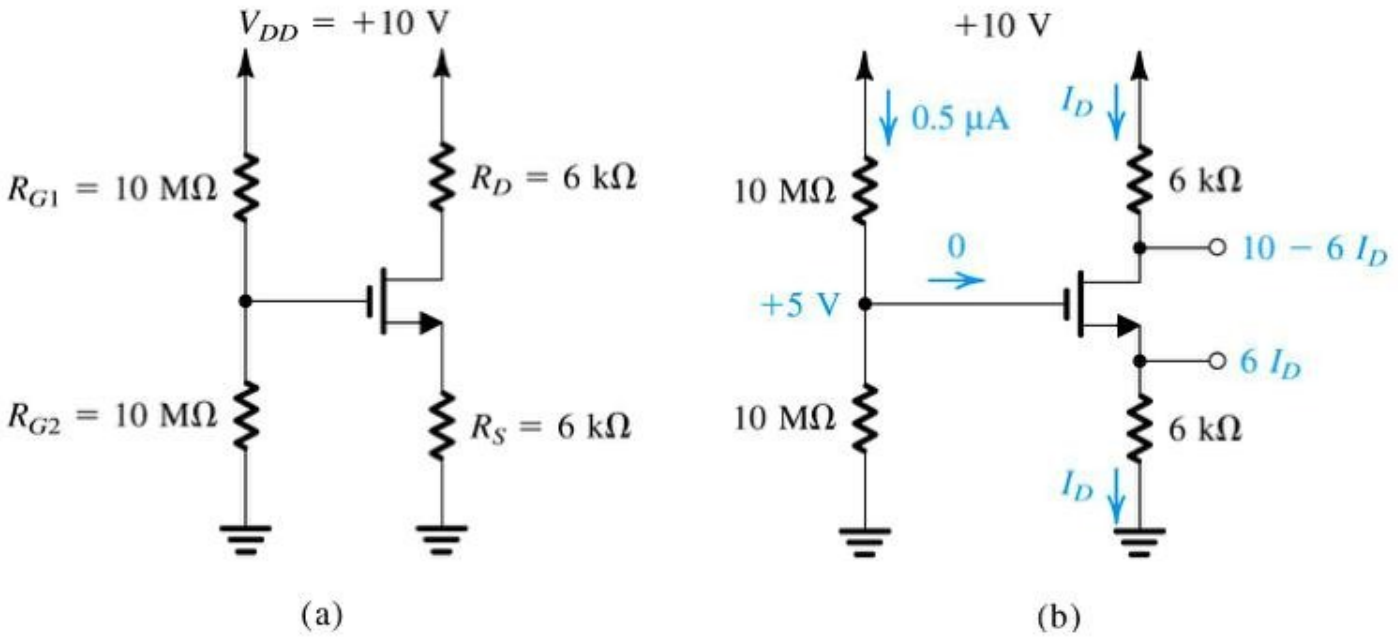


Figure 5.24: (a) Circuit for Example 5.6. **(b)** The circuit with some of the analysis details shown.

Example 5.7: PMOS Transistor

- **Problem Statement:** Design the circuit of Figure 5.25 so that transistor operates in saturation with $I_D = 0.5mA$ and $V_D = +3V$. Let the enhancement-type PMOS transistor have $V_{tp} = -1V$ and $k'_p(W/L) = 1mA/V^2$. Assume $\lambda = 0$.
- **Q:** What is the largest value that R_D can have while maintaining saturation-region operation?

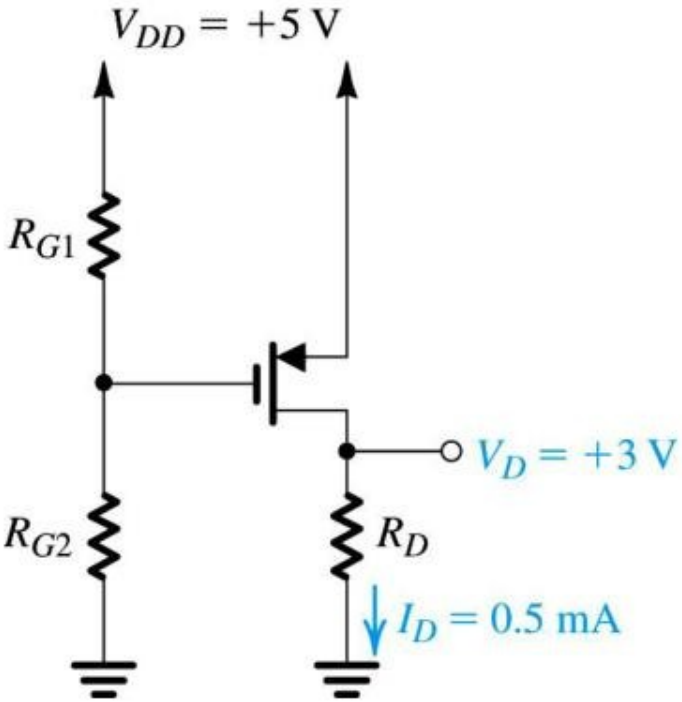


Figure 5.25: Circuit for Example 5.7.

Exercise 5.8: CMOS Transistor

- **Problem Statement:** The NMOS and PMOS transistors in the circuit of Figure 5.26(a) are matched, with $k'_n(W_n/L_n) = k'_p(W_p/L_p) = 1 mA/V^2$ and $V_{tn} = -V_{tp} = 1V$. Assuming $\lambda = 0$ for both devices.
- **Q:** Find the drain currents i_{DN} and i_{DP}, as well as voltage v_O for $v_I = 0V$, +2.5V, and -2.5V.

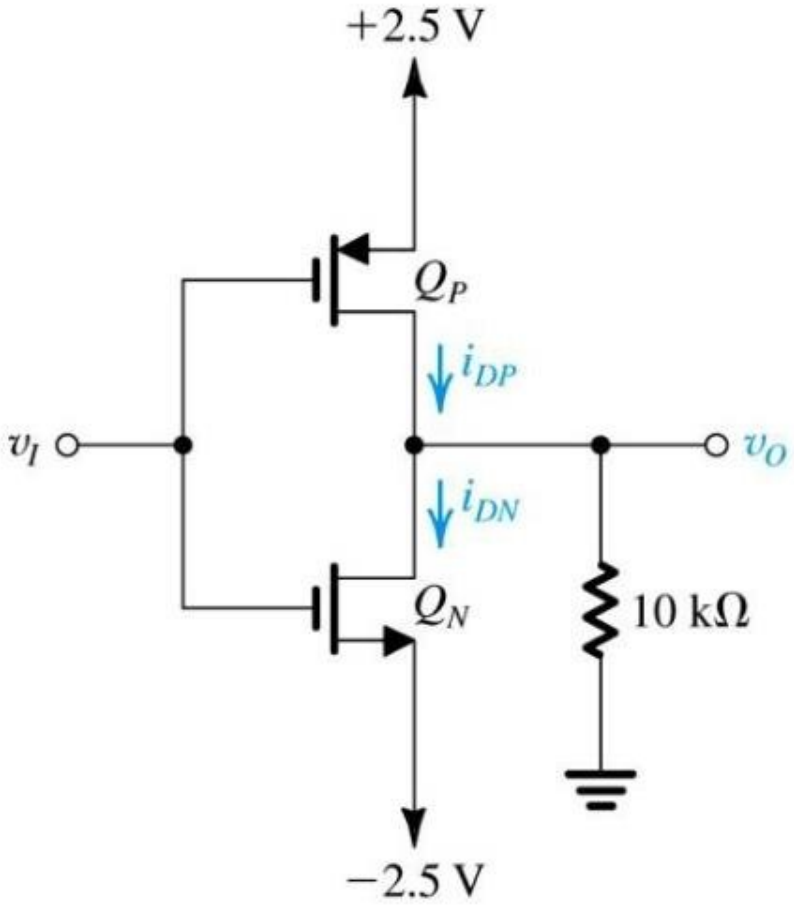


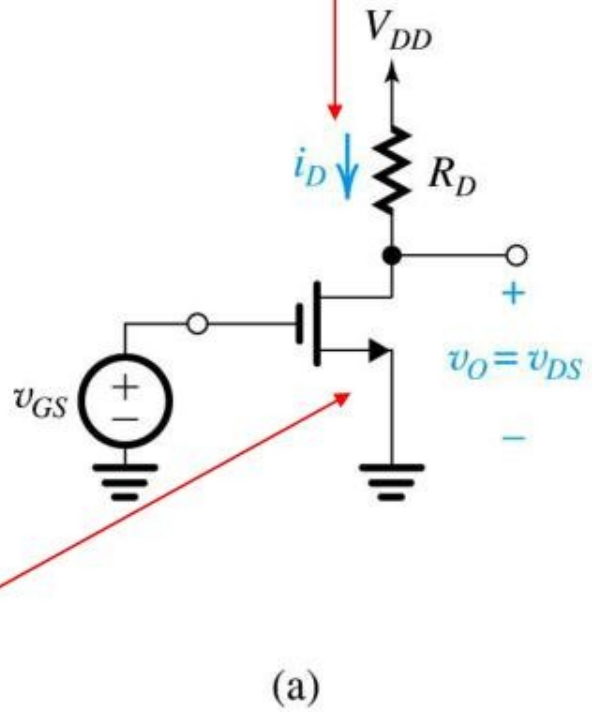
Figure 5.26: Circuits for Example 5.8.

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5.4.1. **Obtaining a Voltage Amplifier**

example of transconductance amplifier

- In section 1.5 of text, we learned that voltage controlled current source (VCCS) can serve as **transconductance amplifier**.
  - the following slides (with blue tint) are a review
- **Q:** How can we translate current output to voltage?
  - **A:** Measure voltage drop across load resistor.

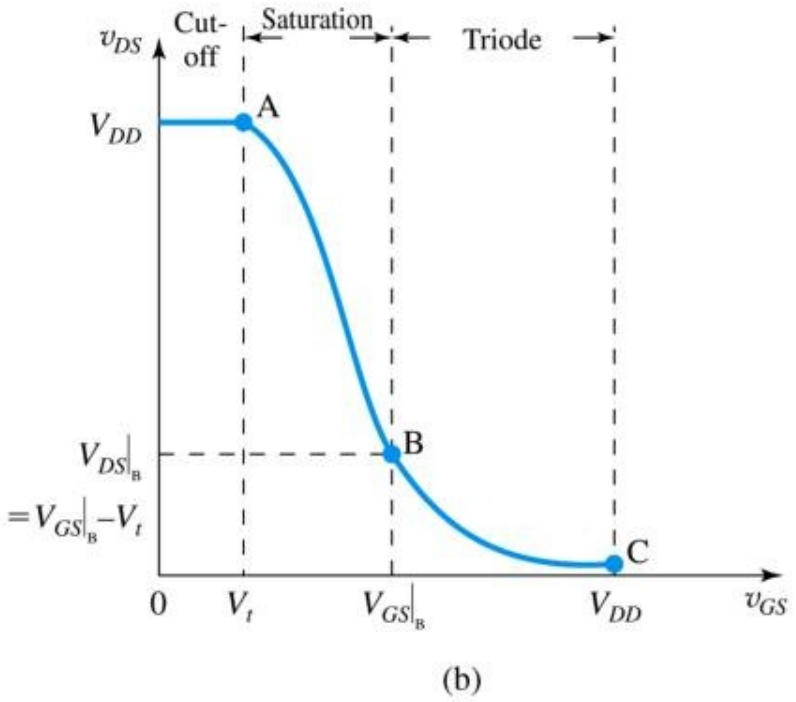


v_{out} supply function of input v_G

(eq5.30) $v_{DS} = v_{DD} - i_D R_D$

Figure 5.27: (a) simple MOSFET amplifier with input v_{GS} and output v_{DS}

5.4.2. Voltage Transfer Characteristic



(b)

Figure 5.27: (b) the voltage transfer characteristic (VTC) of the amplifier from previous slide

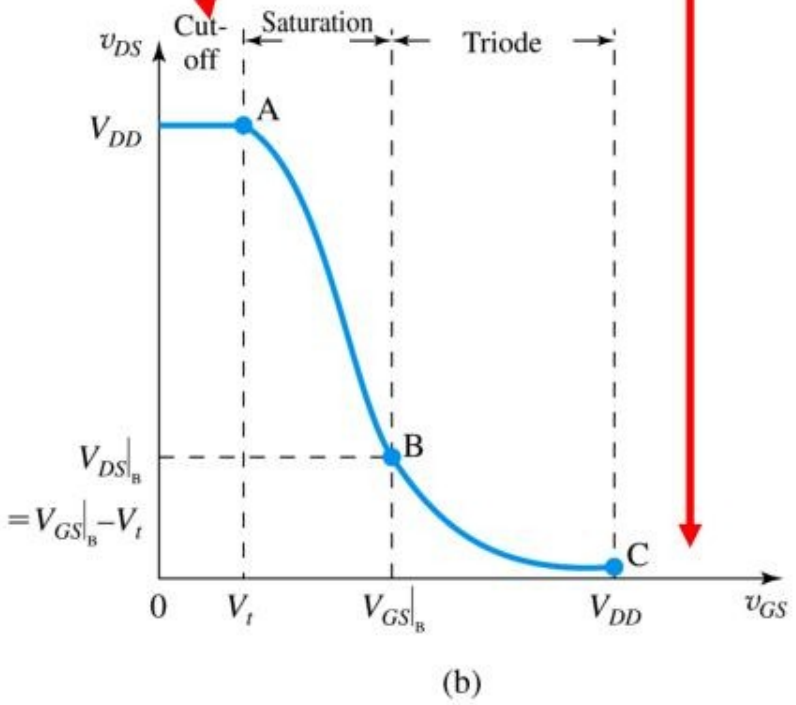
- **voltage transfer characteristics** (VTC) – plot of out voltage vs. input
- **three regions** exist in VTC
  - $v_{GS} < V_t \rightarrow$ **cut off FET**
    - $v_{OV} = v_{GS} - V_t < 0$
    - $I_D = 0$
    - v_{DS} ??? v_{OV}
    - $v_{out} = v_{DD}$
  - $V_t < v_{GS} < v_{DS} + V_t \rightarrow$ **saturation**
    - $v_{OV} = v_{GS} - V_t > 0$
    - $I_D = \frac{1}{2} k_n (v_{GS} - V_t)^2$
    - $v_{DS} >> v_{OV}$
    - $v_{out} = V_{DD} - I_D R_D$
  - $v_{DS} + V_t < v_{GS} < V_{DD} \rightarrow$ **triode**
    - $v_{OV} = v_{GS} - V_t > 0$
    - $I_D = k_n (v_{GS} - V_t - v_{DS}) v_{DS}$
    - $v_{DS} > v_{OV}$
    - $v_{out} = V_{DD} - I_D R_D$

cutoff FET

cutoff AMP

Voltag

Characteristic



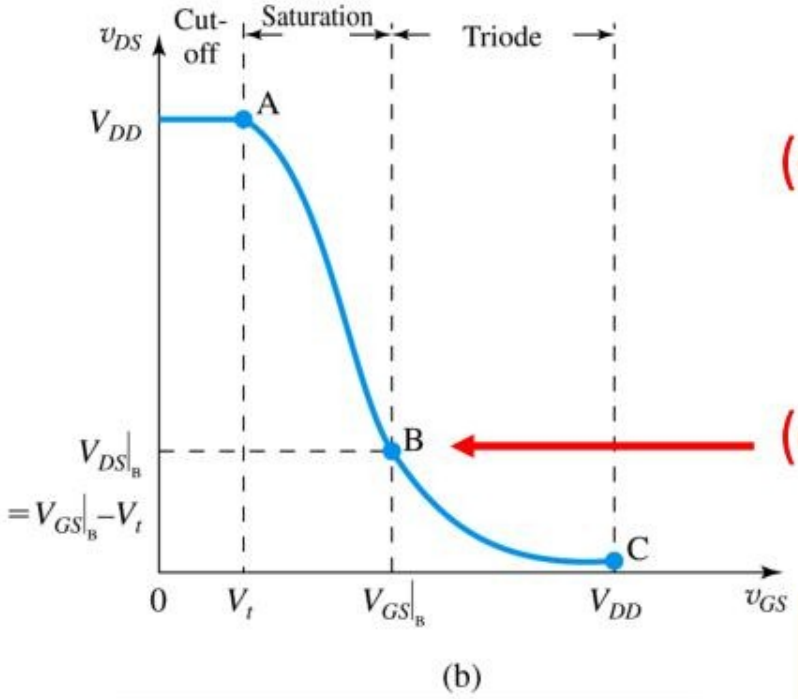
(b)

Figure 5.27: (b) the voltage transfer characteristic (VTC) of the amplifier from previous slide

- **Q:** What observations may be drawn?
  - **A:** Cutoff FET represents transistor blocking, cutoff AMP represents $v_{out} = 0$
  - **A:** As v_{GS} increases…
    - v_{DS} (effectively) decreases
    - i_D increases
    - v_{out} decreases nonlinearly
    - gain (G) decreases
  - **A:** Once $v_{DS} > v_{DD}$, all power is dissipated by resistor R_D

5.4.2. **Voltage Transfer Characteristic**

Q: How do we define v_{DS} in terms of v_{GS} for saturation?



(b)

Figure 5.27: (b) the voltage transfer characteristic (VTC) of the amplifier from previous slide

this is equation is simply ohm's law / KVL

(eq5.32) $$v_{DS} = V_{DD} - \underbrace{\left[\frac{1}{2}k_n\left(v_{GS} - V_t\right)^2\right]}_{i_D} R_D$$

(eq5.33) $$V_{GS}\big|_B = V_t + \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D}$$

Q: How do we define point *B* – boundary between saturation and triode regions?

5.4.3. Biasing the MC to Obtain Linear Amplification

This equation differs from (5.32) because it considers dc component only.

this equation is simply ohm's law

(eq5.34) $$V_{DS} = \underbrace{V_{DD} - \left[\frac{1}{2} k_n \left(V_{GS} - V_t\right)^2\right] R_D}_{V_{source} - I_D R_D}$$

- **Q:** How can we linearize VTC?
  - **A:** Appropriate biasing technique
  - **A:** Dc voltage v_{GS} is selected to obtain operation at point *Q* on segment *AB*
- **Q:** How do we choose v_{GS}?
  - **A:** Will discuss shortly…

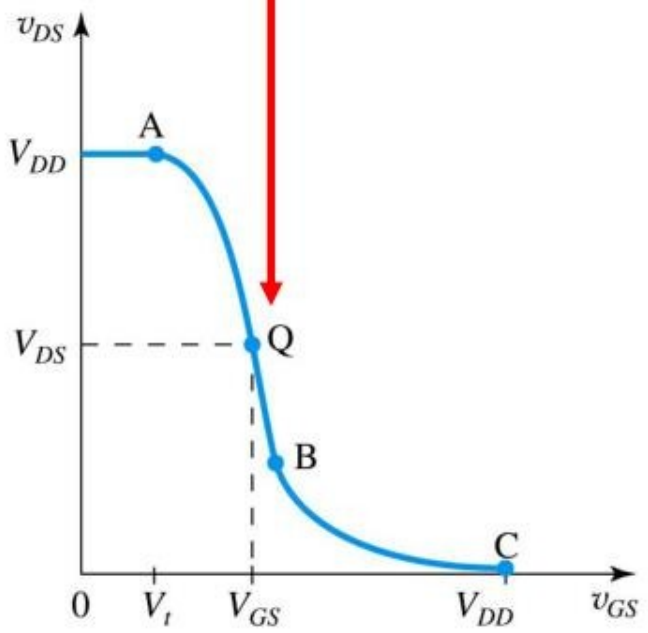


Figure 5.28: biasing the MOSFET amplifier at point *Q* located on segment *AB* of VTC

5.4.3. Biasing the MOSFET to Obtain Linear Amplification

this equation is simply ohm's law

(eq5.34) $$V_{DS} = \underbrace{V_{DD} - \left[\frac{1}{2} k_n \left(V_{GS} - V_t\right)^2\right] R_D}_{V_{source} - I_D R_D}$$

- **bias point / dc operating pt.** (*Q*) – point of linearization for MOSFET
  - Also known as **quiescent point**.
- **Q:** How will *Q* help us?
  - **A:** Because VTC is linear near *Q*, we may perform linear amplification of signal << Q

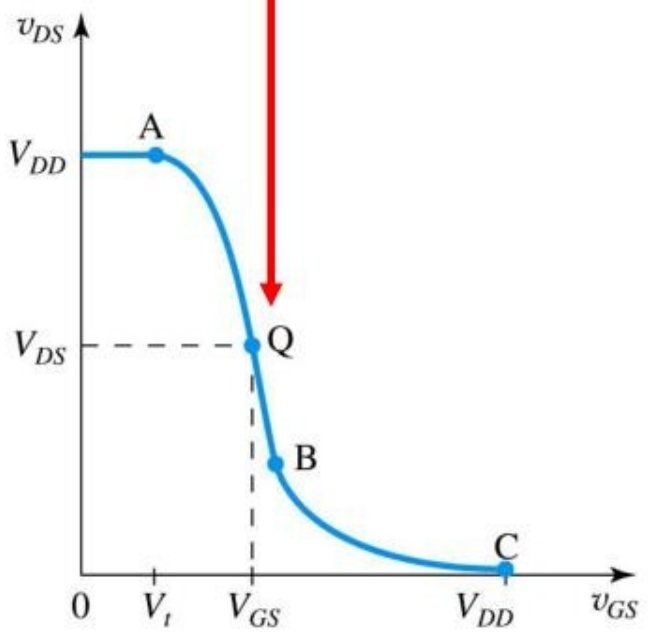
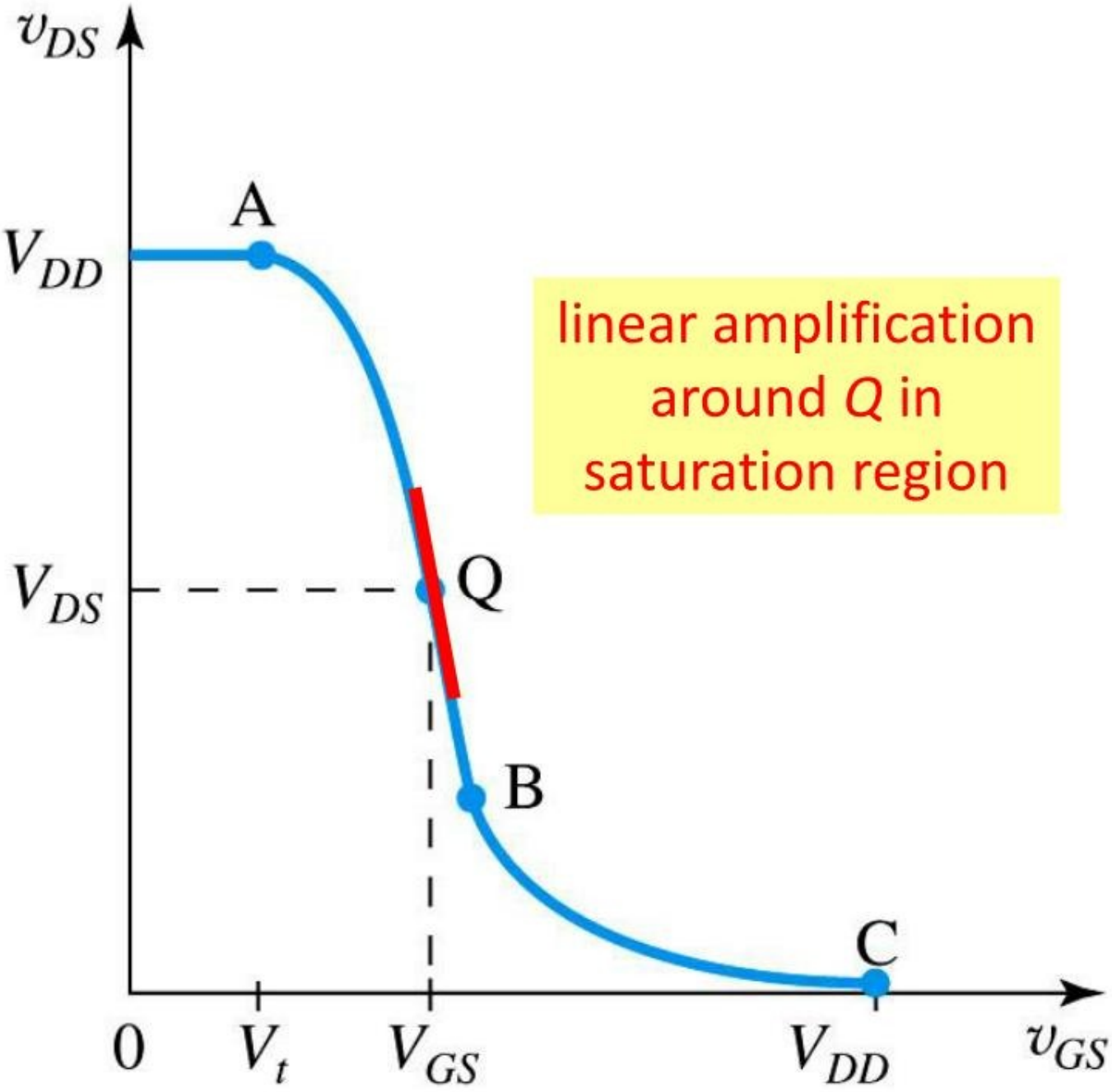


Figure 5.28: biasing the MOSFET amplifier at point *Q* located on segment *AB* of VTC

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v_{DS}
A
V_{DD}
linear amplification
around Q in
saturation region
V_{DS}
Q
B
C
0
V_t
V_{GS}
V_{DD}
v_{GS}

5.4.3. Biasing the MOSFET to Obtain Linear Amplification

- **Q:** How is linear gain achieved?
  - **step #1:** Bias MOSFET with dc voltage V_{GS} as defined by (5.34)
  - **step #2:** Superimpose amplifier input (v_{gs}) upon V_{GS}.
  - **step #3:** Resultant v_{ds} should be linearly proportional to small-signal component v_{gs}.

$$\mathbf{v}_{GS}(t) = V_{GS} + \mathbf{v}_{gs}(t)$$

$$\mathbf{v}_{ds}(t) \propto \mathbf{v}_{gs}(t)$$

Q: How is linear gain achieved?

As long as $\mathbf{v}_{gs}(t)$ is small, its effect on $v_{DS}(t)$ will be linear – *facilitating linear amplification.*

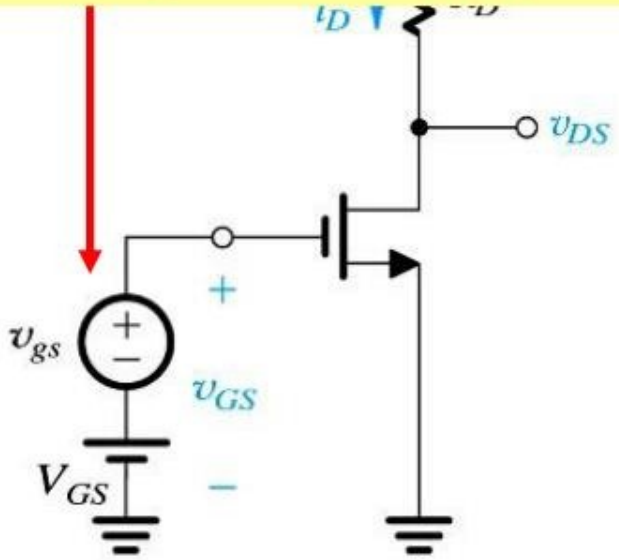
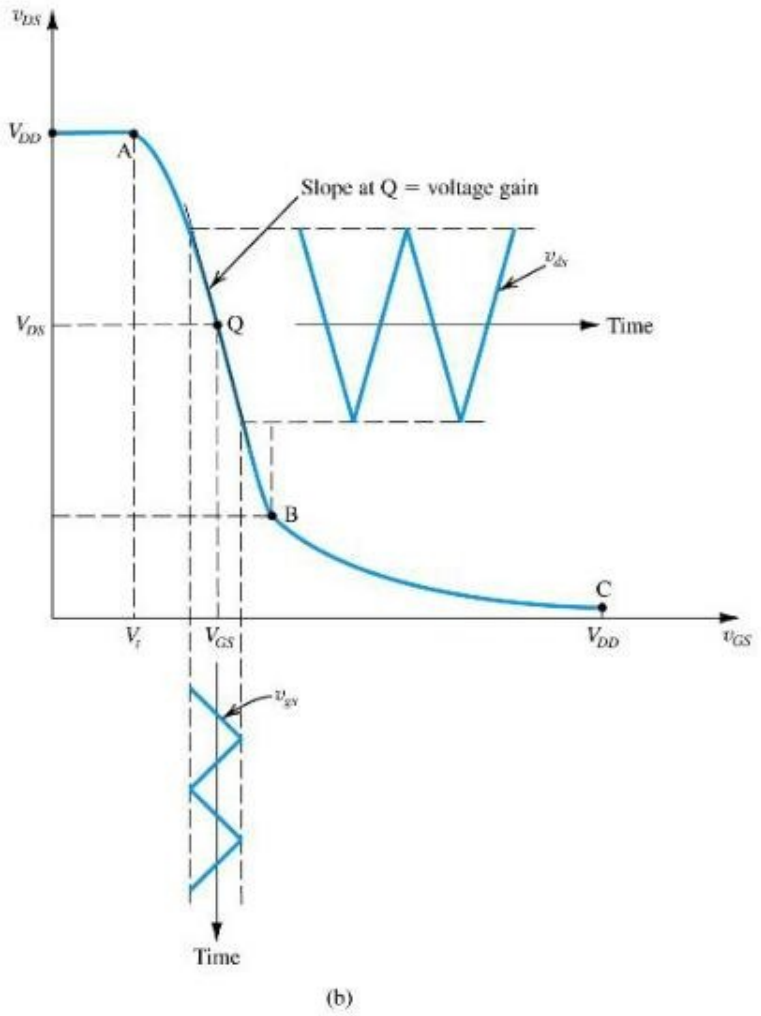


Figure 5.29: The MOSFET amplifier with a small time-varying signal $v_{gs}(t)$ superimposed on the dc bias voltage v_{GS}. The MOSFET operates on a short almost-linear segment of the VTC around the bias point Q and provides an output voltage $v_{ds} = A_v v_{gs}$



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Q: How is linear gain achieved?

- **step #4:** Note if v_{gs} is small, output v_{ds} will be nearly linearly proportional to it.
  - Slope will be constant.

$$\text{(eq5.35)}\quad A_v = \left.\frac{dv_{DS}}{dv_{GS}}\right|_{\underbrace{v_{GS}\approx V_{GS}}_{\text{means that } v_{gs} \text{ is small}}}$$

action: replace v_{DS} with (5.32)

$$\text{(eq5.35)}\quad A_v = \left.\frac{d\left(V_{DD} - \frac{1}{2}k_n\left(v_{GS} - V_t\right)^2 R_D\right)}{dv_{GS}}\right|_{v_{GS}\approx V_{GS}}$$

action: simplify

$$\text{(eq5.36)}\quad A_v = \overbrace{-k_n\left(V_{GS} - V_t\right)R_D}$$

action: replace with V_{OV}

$$\text{(eq5.37)}\quad A_v = \overbrace{-k_n V_{OV} R_D}$$

5.4.4. Small-Signal Voltage Gain

- **Q:** What observations can be made about voltage gain?
  - **A:** Gain is negative.
  - **A:** Gain is proportional to:
    - load resistance (R_D)
    - transistor conductance parameter (k_n)
    - overdrive voltage (v_{OV})

$$\text{(eq5.35)}\quad A_v = \left.\frac{dv_{DS}}{dv_{GS}}\right|_{\underbrace{v_{GS}\approx V_{GS}}_{\text{means that } v_{gs} \text{ is small}}}$$

$$\text{(eq5.35)}\quad A_v = \overbrace{\left.\frac{d\left(V_{DD} - \frac{1}{2}k_n\left(v_{GS} - V_t\right)^2 R_D\right)}{dv_{GS}}\right|_{v_{GS}\approx V_{GS}}}^{\textbf{action: replace } v_{DS} \text{ with (5.32)}}$$

$$\text{(eq5.36)}\quad A_v = \overbrace{-k_n\left(V_{GS} - V_t\right)R_D}^{\textbf{action: simplify}}$$

$$\text{(eq5.37)}\quad A_v = \overbrace{-k_n V_{OV} R_D}^{\textbf{action: replace with } V_{OV}}$$

5.4.4. Small-Signal Gain

- Equation (5.38) is another version of (5.37) which incorporates (5.17).
- It demonstrates that gain is ratio of:
  - voltage drop across R_D
  - half of over voltage

(eq5.37) $A_v = -k_n V_{OV} R_D$

action: incorporate (5.17) $i_D = \frac{1}{2} k_n v_{OV}^2$

(eq5.38) $A_v = -\left(\frac{I_D R_D}{V_{OV}/2} \right)$

5.4.4. **Small-Signal Gain**

This does not mean that output may be 10x supply (V_{DD}).

For example, 0.13*mm* CMOS technology with *VDD* = 1.3*V* yields maximum gain of 13*V/V*.

- **Q:** How does (5.38) relate to pl
  - **A:** For modern CMOS techn
  than 0.2*V*.
  - **A:** This means that max achievable gain is approximately $10V_{DD}$.

$$\max(A_v) = -\left(\frac{\overbrace{\max(I_D)R_D}^{V_{DD}}}{\underbrace{V_{OV}/2}_{0.1V}}\right) = -10V_{DD}$$

Example 5.9:
MOSFET Amplifier

- **Problem Statement:** Consider the amplifier circuit shown in Figure 5.29(a). The transistor is specified to have $V_t = 0.4V$, $k'_n = 0.4mA/V^2$, $W/L = 10$, and $\lambda = 0$. Also, let $V_{DD} = 1.8V$, $R_D = 17.5kOhms$, and $V_{GS} = 0.6V$.
- **Q(a):** For $v_{gs} = 0$ (and hence $v_{ds} = 0$), find V_{OV}, I_D, V_{DS}, and A_v.
- **Q(b):** What is the maximum symmetrical signal swing allowed at the drain? Hence, find the maximum allowable amplitude of a sinusoidal v_{gs}.

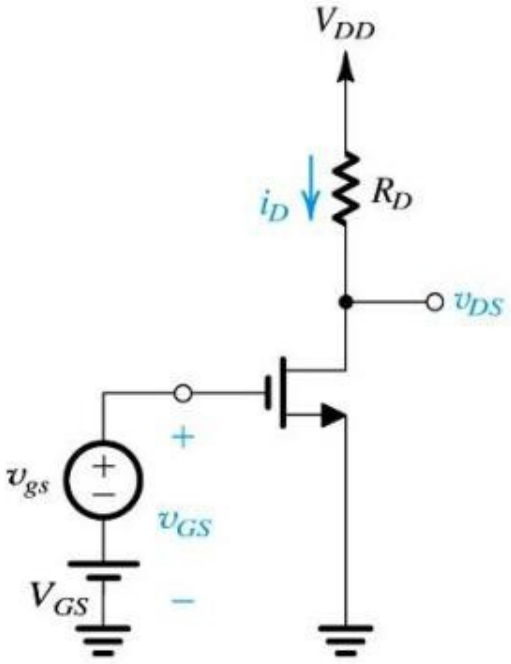


Figure 5.29:

5.4.5. Determining the VTC via Graphical Analysis

(eq5.39) $$i_D = \frac{V_{DD}}{R_D} - \frac{v_{DS}}{R_D}$$

- Graphical method for determining VTC is shown in Figure 5.31
- Rarely used in practice, b/c difficult to draw *vi*-relationship.
- Based on observation that, for each value of v_{GS}, circuit will operate at intersection of i_D and v_{DS}.

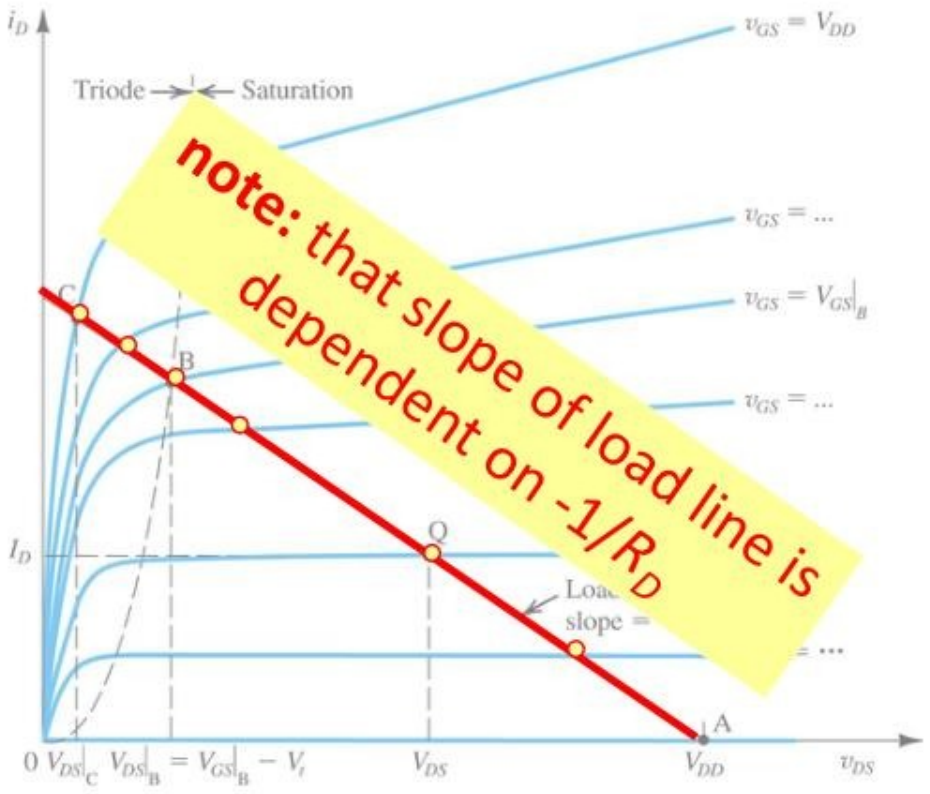
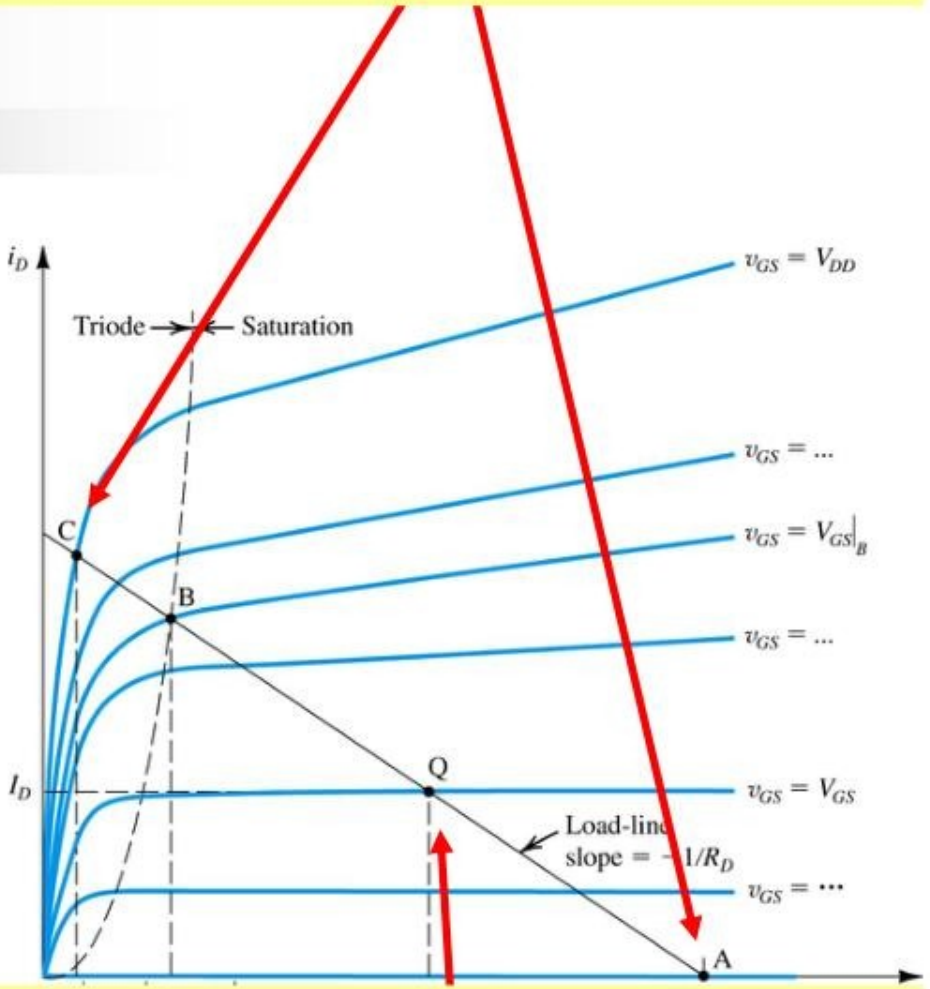


Figure 5.31: Graphical construction to determine the voltage transfer characteristic of the amplifier in Fig. 5.29(a).

5.4.5. Determining the VTC via Graphical Analysis

- point A – where $v_{GS} = V_t$
- point Q – where MOSFET may be biased for amplifier operation
  - $v_{GS} = V_{GS}$, $v_{DS} = V_{DS}$
- point B – where MOSFET leaves saturation / enters triode
- point C – where MOSFET is deep in triode region and $v_{GS} = V_{DD}$

Points *A* (open) and *C* (closed) are suitable for switch applications



Point *Q* is suitable for amplifier applications

5.4.5. Determining the VTC via Graphical Analysis

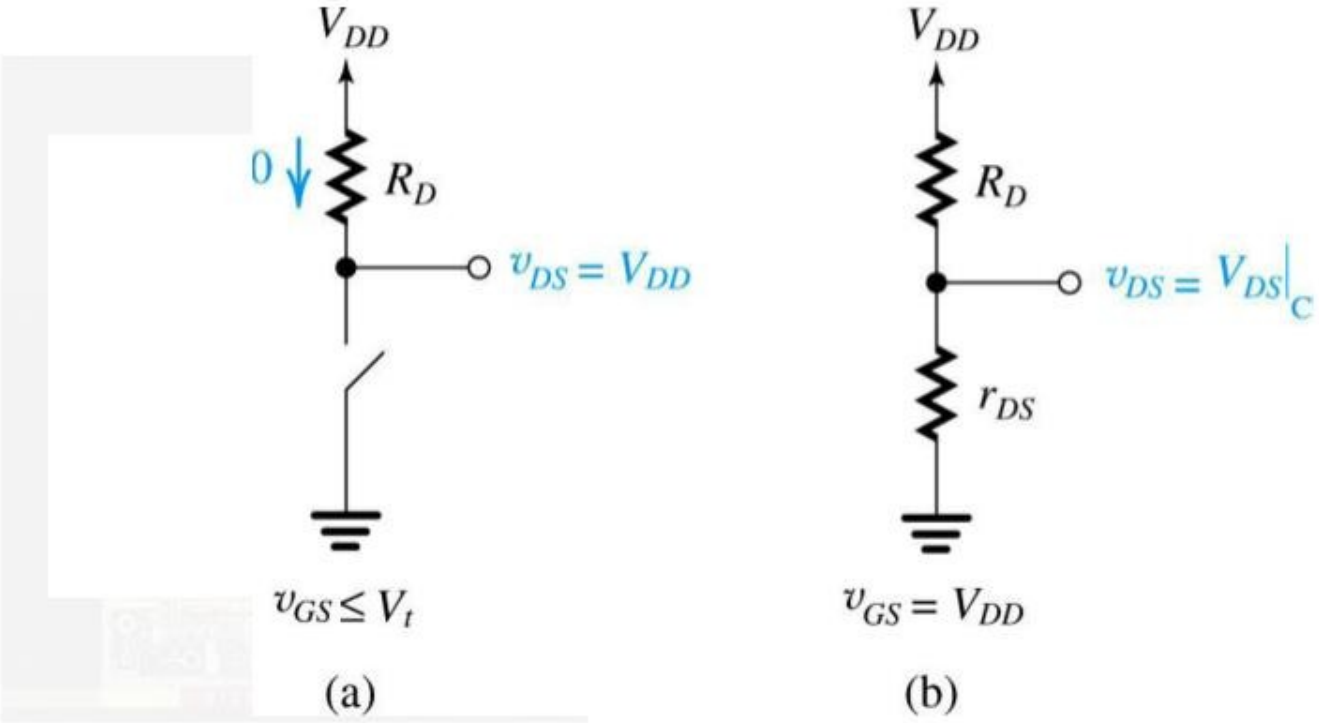


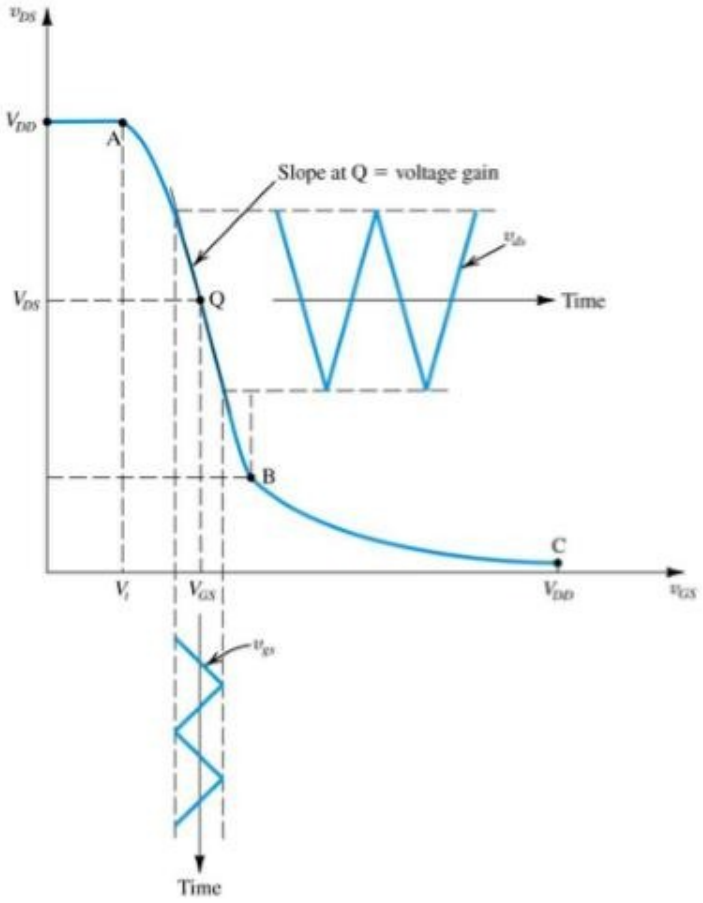
Figure 5.32: Operation of the MOSFET in Figure 5.29(a) as a switch: **(a)** Open, corresponding to point A in Figure 5.31; **(b)** Closed, corresponding to point C in Figure 5.31. The closure resistance is approximately equal to r_{DS} because V_{DS} is usually very small.

5.4.6. **Locating the Bias Point Q**

- **bias point** (Q) – is determined by value of v_{GS} and load resistance R_D.
- Two considerations in deciding Q:
  - Required gain.
  - Allowable signal swing at output.

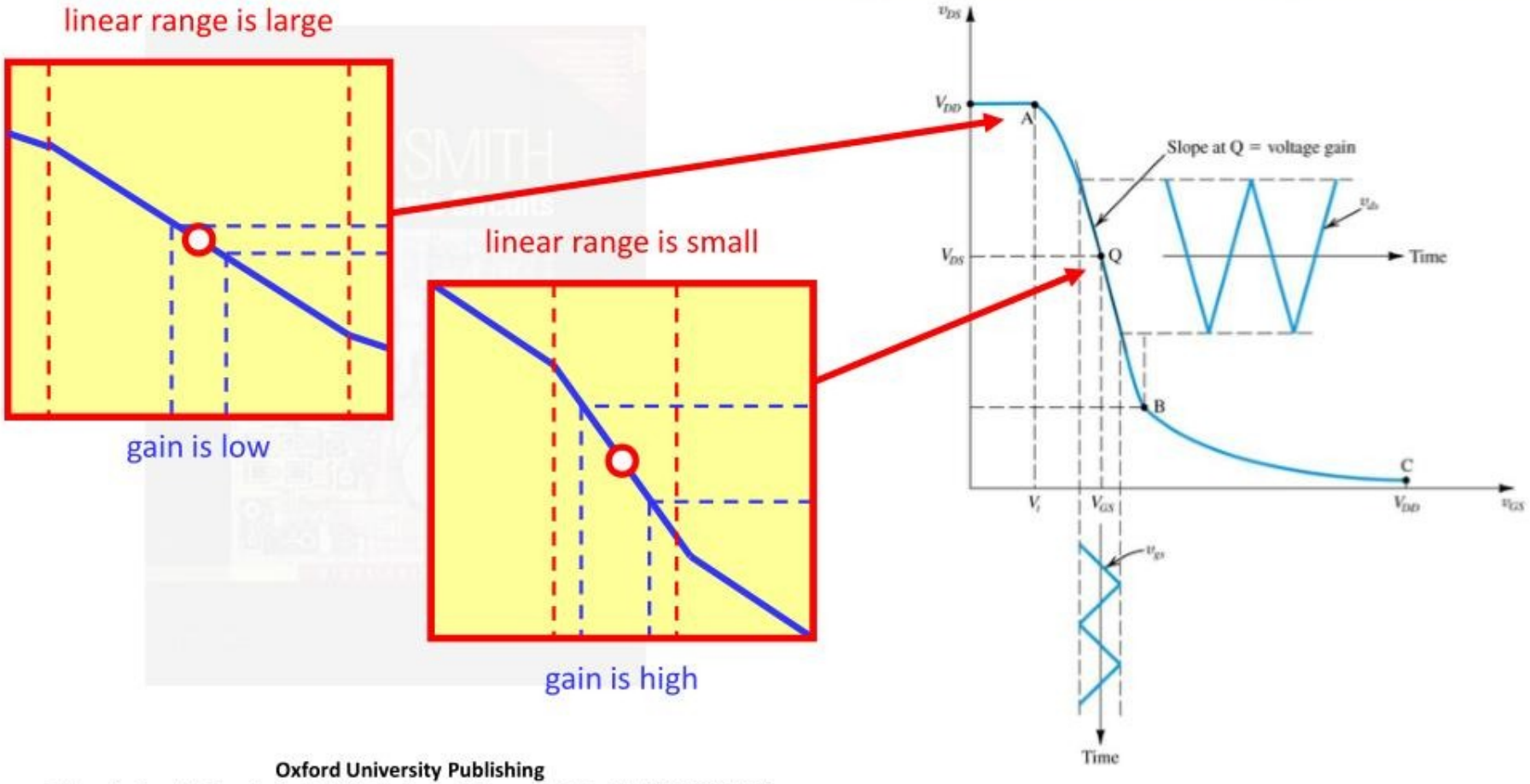
5.4.6. **Locating the Bias Point Q**

- **Q:** How is *Q* for VTC defined (assuming R_D is fixed)?
  - A: As point *Q* approaches *B*:
    - gain increases
    - maximum v_{gs} swing decreases



5.4.6. **Locating the Bias Point Q**

Note that a trade-off between gain and linear range exists.



5.4.6. Locating the Bias Point Q

The objective is to prevent v_{DS} from "clipping" or entering triode region

- To define **load resistance R_D**, one should refer to the i_D - v_{DS} plane.
- Two examples of R_D are shown to right for illustration:
  - **Q2:** too close to triode
    - not enough **legroom**
  - **Q1:** too close to V_{DD}
    - not enough **headroom**
- Ideally, we want to be somewhere in the middle.

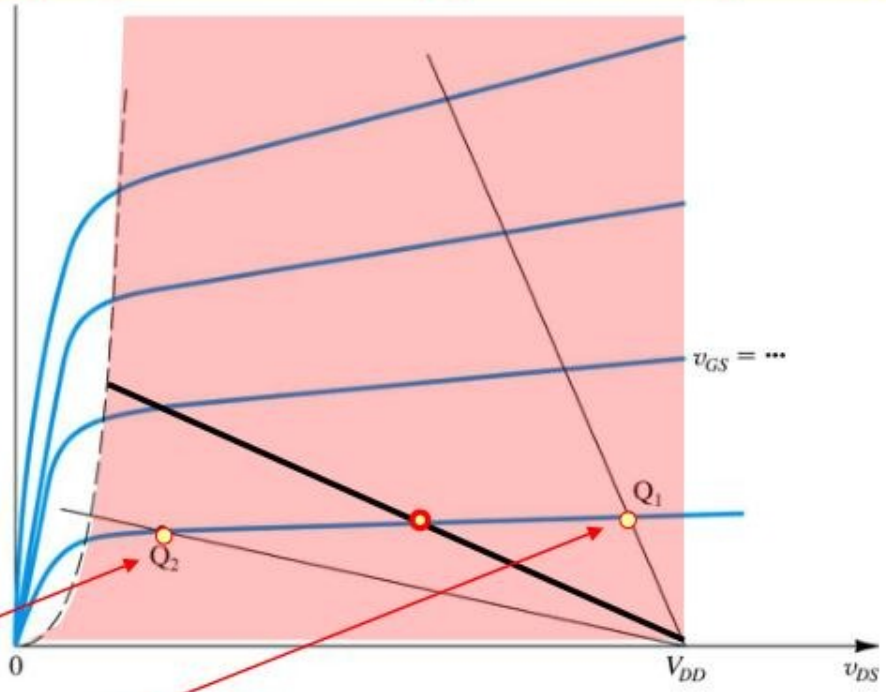


Figure 5.33: Two load lines and corresponding bias points. Bias point Q_1 does not leave sufficient room for positive signal swing at the drain (too close to V_{DD}). Bias point Q_2 is too close to the boundary of the triode region and might not allow for sufficient negative signal swing.

5.5. S
Ope
Models

input voltage to be amplified

dc bias voltage

output voltage

- Previously it was stated that linear amplification may be obtained from MOSFET via…
  - Operation in saturation region
  - Utilization of small-input
- This section will explore small-signal operation in detail
  - Note the conceptual amplifier circuit to right

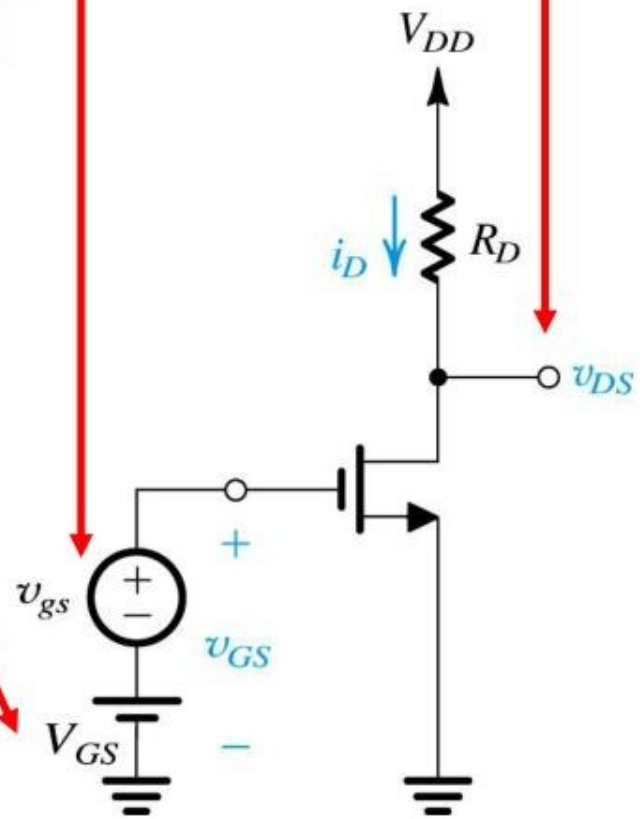


Figure 5.34: Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

5.5.1. The DC Bias Point

- **Q:** How is dc bias current I_D defined?

only applies in saturation where $V_{DS} > V_{OV}$

(eq5.40) $$I_D = \frac{1}{2} k_n \left(V_{GS} - V_t\right)^2 = \frac{1}{2} k_n V_{OV}^2$$

(eq5.41) $$V_{DS} = V_{DD} - R_D I_D$$

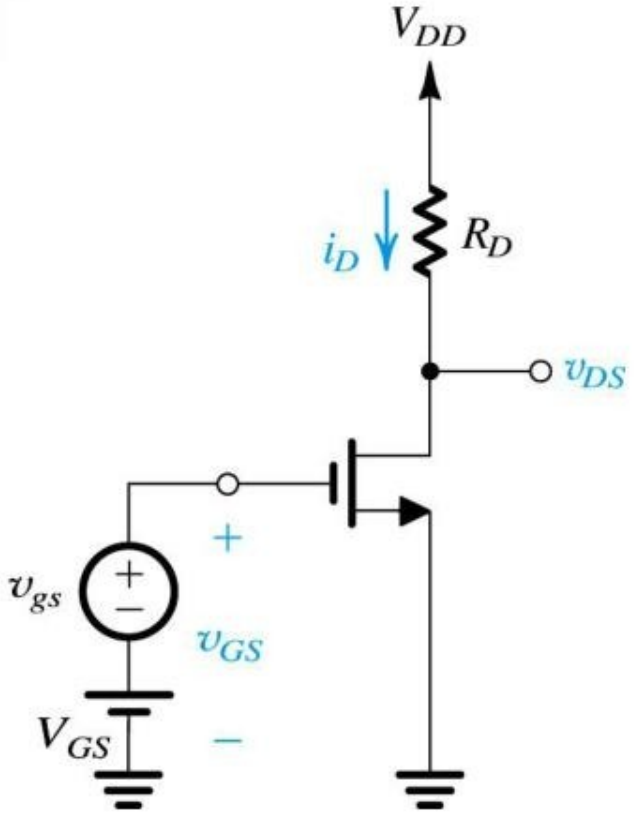


Figure 5.34: Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

5.5.2. The Signal Current in the Drain Terminal

- **Q:** What is effect of v_{gs} on i_D?
  - **step #1:** Define v_{GS} as in (5.42).
  - **step #2:** Define i_D, separate terms as function of V_{GS} and v_{gs}

(eq5.42) $v_{GS} = V_{GS} + v_{gs}$

(eq5.17) $i_D = \frac{1}{2}k_n\left(\underbrace{\underbrace{V_{GS}+v_{gs}}_{v_{GS}} - V_t}_{v_{OV}}\right)^2$ — **action:** state (5.17)

(eq5.43) $i_D = \frac{1}{2}k_n\underbrace{\left[\left(V_{GS}-V_t\right)^2 + \ldots \; \ldots + 2\left(V_{GS}-V_t\right)v_{gs} + v_{gs}^2\right]}_{\left(V_{GS}+v_{gs}-V_t\right)}$ — **action:** expand the squared term via $V_{GS}-V_t$ and v_{gs}

$i_D = \frac{1}{2}k_n\left(V_{GS}-V_t\right)^2 + \ldots$

$\ldots + k_n\left(V_{GS}-V_t\right)v_{gs} + \frac{1}{2}k_n v_{gs}^2$ — **action:** simplify

Note that this differs from previous analyses - because of attempt to isolate the effect of v_{gs} from V_{GS}.

Q: What is effect of v_{gs} on i_D?

Note that to minimize nonlinear distortion, v_{gs} should be kept small.

$$\tfrac{1}{2}k_n v_{gs}^2 \ll k_n(V_{GS}-V_t)v_{gs}$$

$$v_{gs} \ll 2(V_{GS}-V_t)$$

$$v_{gs} \ll 2v_{OV}$$

- **step #3:** Classify terms.
  - dc bias current (I_D).
  - linear gain – is desirable.
  - nonlinear distortion – is undesirable, because rep. distortion.

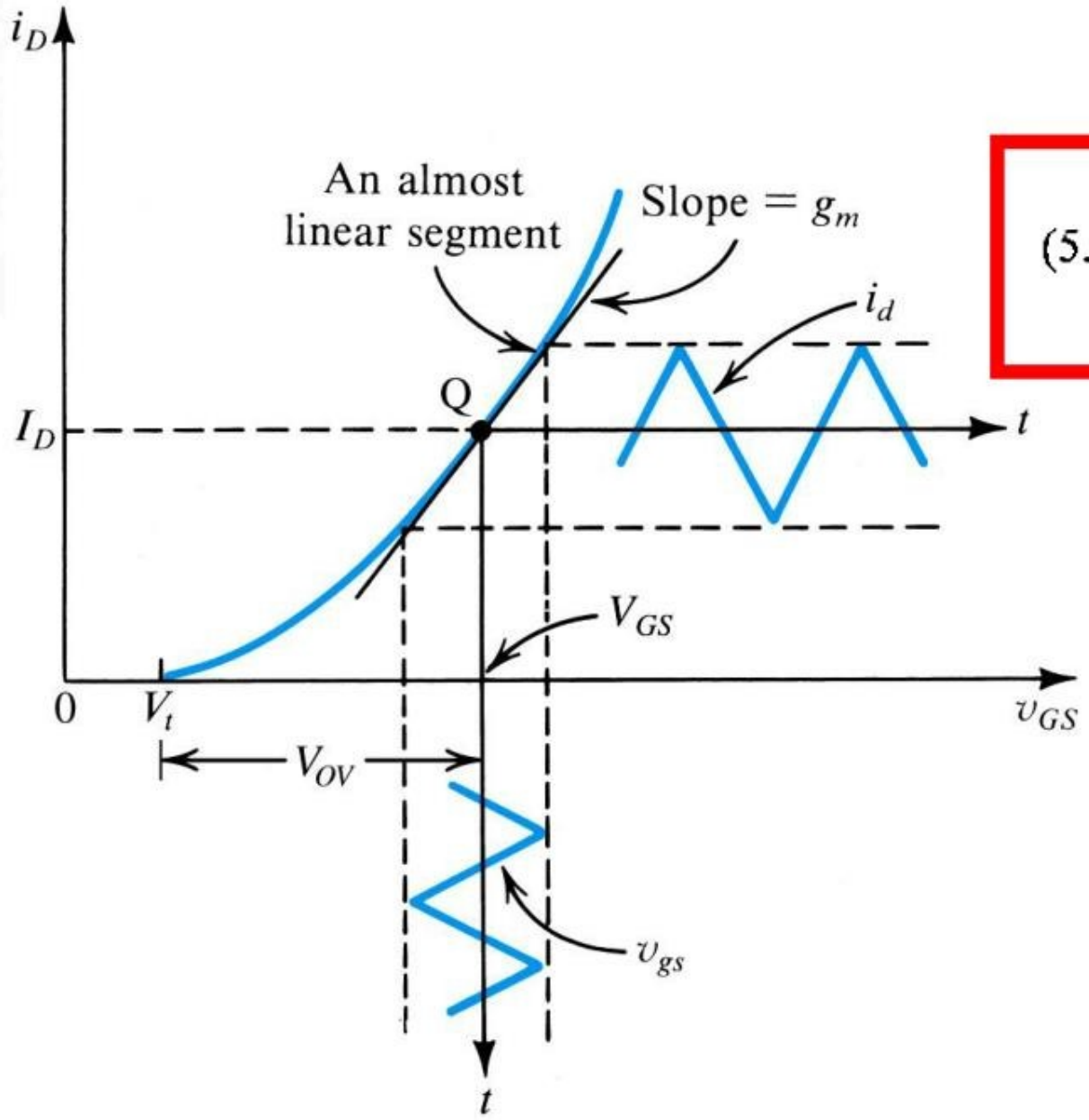
(eq5.43) $$i_D = \underbrace{\frac{1}{2}k_n\left(V_{GS}-V_t\right)^2}_{\text{dc bias current } (I_D)} + \underbrace{k_n\left(V_{GS}-V_t\right)v_{gs}}_{\text{linear gain term}} + \underbrace{\frac{1}{2}k_n v_{gs}^2}_{\text{nonlinear distortion term}}$$

Q: What is effect of v_{gs} on i_D?

- **step #4:** Adapt (5.43) for small-signal condition.
  - If $v_{gs} << 2v_{OV}$, neglect distortion.

(eq5.43) $$i_D = \underbrace{\frac{1}{2}k_n\left(V_{GS}-V_t\right)^2}_{\text{dc bias current } (I_D)} + \underbrace{k_n\left(V_{GS}-V_t\right)v_{gs}}_{\substack{\text{linear} \\ \text{gain} \\ \text{term}}} + \underbrace{\frac{1}{2}k_n v_{gs}^2}_{\substack{\text{nonlinear} \\ \text{distortion} \\ \text{term}}}$$

(eq5.47) **MOSFET transconductance** $$g_m = \frac{v_{gs}}{i_d} = k_n\left(V_{GS}-V_t\right)$$



$$(5.49) \qquad g_m \equiv \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{v_{GS}=V_{GS}}$$

Figure 5.35: Small-signal operation of the MOSFET amplifier.

5.5.3. The Voltage Gain

- **Q:** How is voltage gain (A_v) defined?
  - **step #1:** Define v_{DS} for circuit of Figure 5.34 using KVL.

$$v_{DS} = V_{DD} - R_D i_D = \overbrace{V_{DD} - R_D\left(I_D + i_d\right)}^{\text{action: apply small-signal condition}}$$

$$v_{DS} = \overbrace{V_{DD} - R_D I_D - R_D i_d}^{\text{action: regroup terms}} = \overbrace{V_{DS} - R_D i_d}^{\text{action: simplify}}$$

V_{DS}: dc component (V_{DS}); v_{ds}

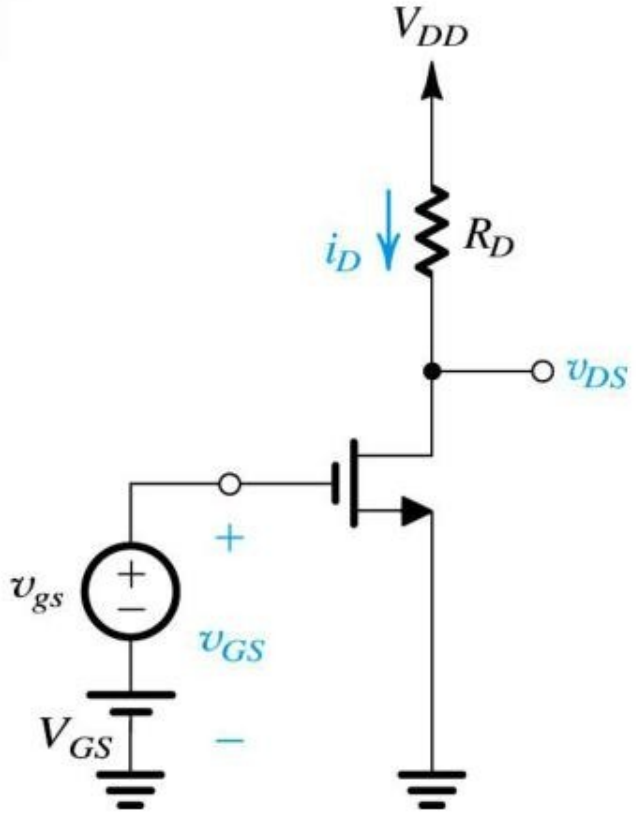


Figure 5.34: Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

Q: How is voltage gain (A_v) defined?

- **step #2:** Isolate v_{ds} component of v_{DS}.
- **step #3:** Solve for gain (A_v).

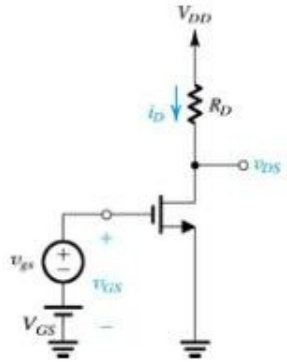


Figure 5.34: Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

action: isolate v_{ds}

$$\text{(eq5.50)}\quad v_{ds} = -R_D i_d$$

action: insert (5.47)

$$\text{(eq5.50)}\quad v_{ds} = -R_D \underbrace{\left(g_m v_{gs}\right)}_{(5.47)}$$

action: solve for gain

$$\text{(eq5.51)}\quad A_v = \frac{v_{ds}}{v_{gs}} = -g_m R_D$$

5.5.3. The Voltage Gain

- Output signal is shifted from input by 180°.
- Input signal $v_{gs} << 2(V_{GS} - V_t)$.
- Operation should remain in MOSFET saturation region
  - $v_{DS} > v_{GS} - V_t$ (legroom)
  - $v_{DS} < V_{DD}$ (headroom)

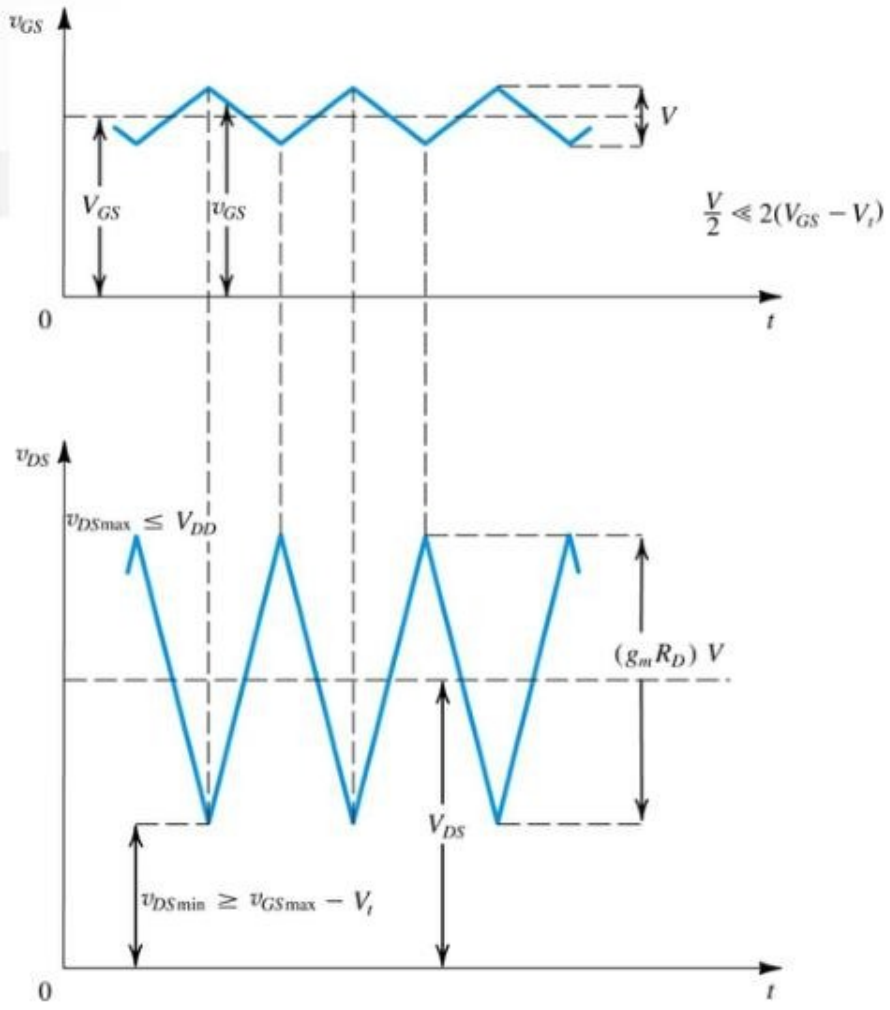


Figure 5.36: Total instantaneous voltage v_{GS} and v_{DS} for the circuit in Figure 5.34.

5.5.5. Small-Signal Equivalent Models

- From signal POV, FET behaves as VCCS.
  - Accepts v_{gs} between gate and source
  - Provides current (i_D) at drain
  - Input resistance is high
    - b/c gate terminal draws $i_G = 0$
  - Output resistance is high

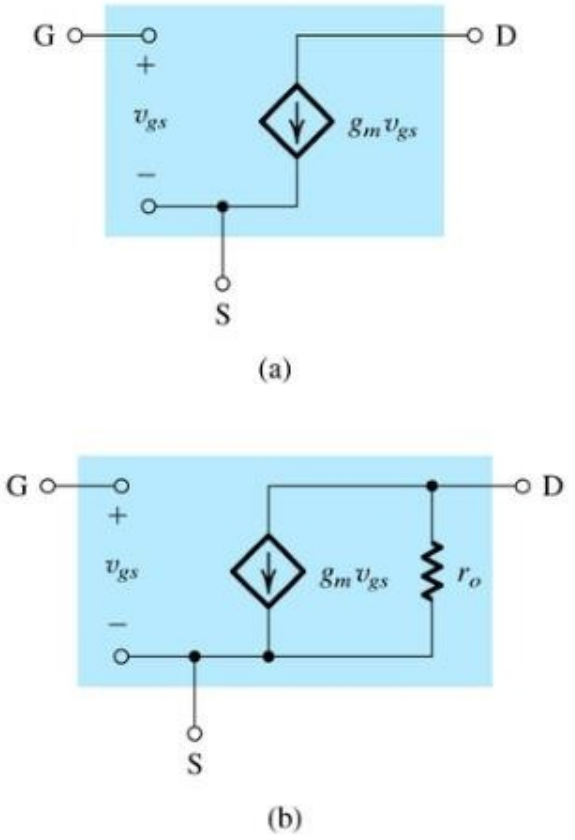


Figure 5.37: Small-signal models for the MOSFET: **(a)** neglecting the dependence of i_D on v_{DS} in saturation (the channel-length modulation effect) and **(b)** including the effect of channel length modulation

5.5.5. **Small-Signal Equivalent Models**

Note that this resistor (r_o) takes on value 10*kOhm* to 1*MOhm* and represents channel-length modulation.

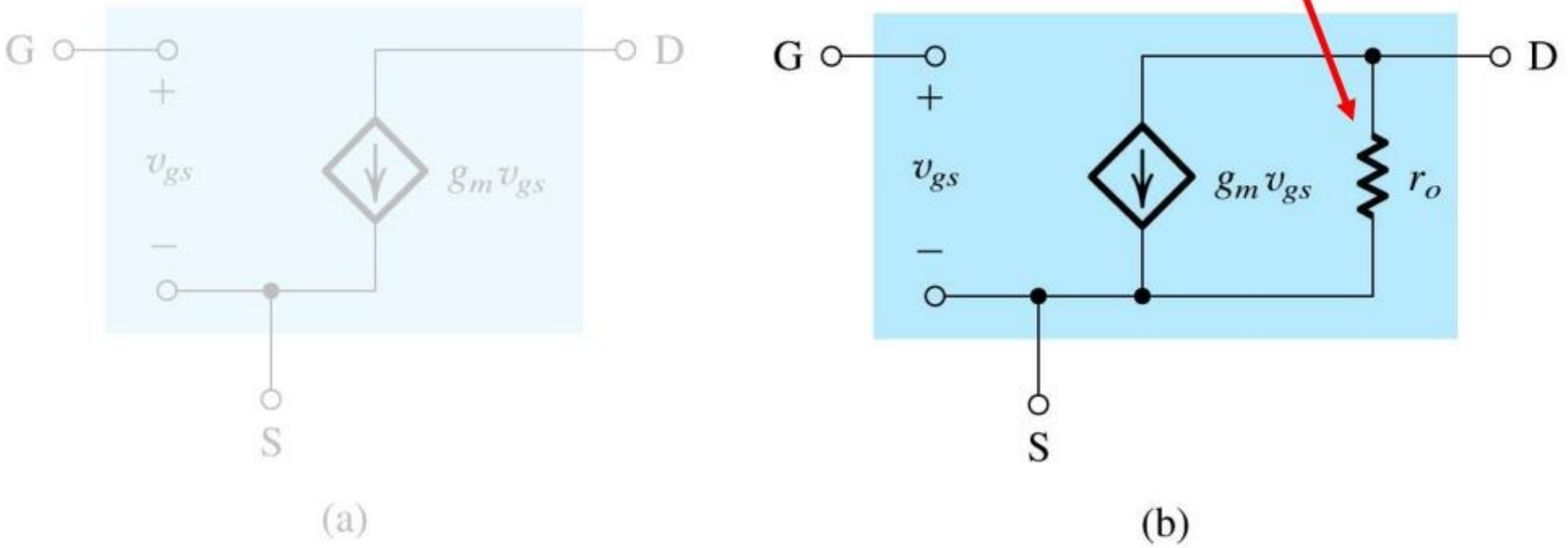


Figure 5.37: Small-signal models for the MOSFET: **(a)** neglecting the dependence of i_D on v_{DS} in saturation (the channel-length modulation effect) and **(b)** including the effect of channel length modulation

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More Observations

- Model (b) is more accurate than model (a)
- $r_o = V_A / I_D$
- Small signal parameters (g_m, r_o) both depend on dc bias point
- If channel-length modulation is considered, (5.51) becomes (5.54).

less accurate, b/c does not consider channel length modulation

(eq5.51) $$A_v = \frac{v_{ds}}{v_{gs}} = -g_m R_D$$

(eq5.54) $$A_v = \frac{v_{ds}}{v_{gs}} = -g_m \left(R_D \,||\, r_o \right)$$

more accurate, b/c does consider channel length modulation

5.5.6. The Transconductance g_m

- Observations from (5.47)
  - g_m is proportional to μ_n, C_{ox}, ratio W/L, dc component V_{OV}.
  - MOSFET with short / wide channel provides maximum gain.
  - Gain may be increased via V_{GS}, but not without reducing allowable swing of v_{gs}.

(eq5.47) $$g_m = \frac{v_{gs}}{i_d} = k_n\left(V_{GS} - V_t\right)$$

action: make some substitutions

(eq5.47) $$g_m = k_n' \frac{W}{L}\left(V_{GS} - V_t\right)$$

k_n

action: simplify

(eq5.55) $$g_m = k_n' \frac{W}{L} V_{OV}$$

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5.5.6: The Transconductance g_m

- Observations from (5.47)
  - g_m is proportional to square root of dc bias current (I_D)
  - For given I_D, g_m is proportional to $(W/L)^{1/2}$
- This behavior is sharp contrast to the bipolar junction transistor (BJT).
  - For which, g_m is proportional to g_m alone (not size or geometry).

(eq5.40) $$I_D = \frac{1}{2} k_n' \frac{W}{L} V_{OV}^2$$

action: solve (5.40) for V_{OV}

$$V_{OV} = \frac{\sqrt{2I_D}}{\sqrt{k_n'}\sqrt{W/L}}$$

(eq5.55) $$g_m = k_n' \frac{W}{L} V_{OV}$$

action: substitute for V_{OV} as defined above

(eq5.56) $$g_m = k_n' \frac{W}{L} \sqrt{\frac{2I_D}{k_n' W/L}}$$

action: simplify

(eq5.56) $$g_m = \sqrt{2k_n'}\sqrt{W/L}\sqrt{I_D}$$

5.5.6. The Transconductance g_m

- **Q:** How does MOSFET compare to BJT? Assume I_D = 0.5mA, k'_n = 120mA/V^2.
  - **A:** MOSFET g_m = 0.35mA/V
    - W/L = 1
  - **A:** MOSFET g_m = 3.5mA/V
    - W/L = 100
  - **A:** BJT g_m = 20mA/V

5.5.6: The Transconductance g_m

- Figure 5.38 illustrates the relationship defined in (5.57).

(eq5.55) $$g_m = k_n' \frac{W}{L} V_{OV}$$

action: replace $k_n' \frac{W}{L}$

(eq5.56) $$g_m = \left(\frac{2I_D}{(V_{GS} - V_t)^2} \right) V_{OV}$$

action: simplify

(eq5.57) $$g_m = \frac{2I_D}{V_{GS} - V_t} = \frac{2I_D}{V_{OV}}$$

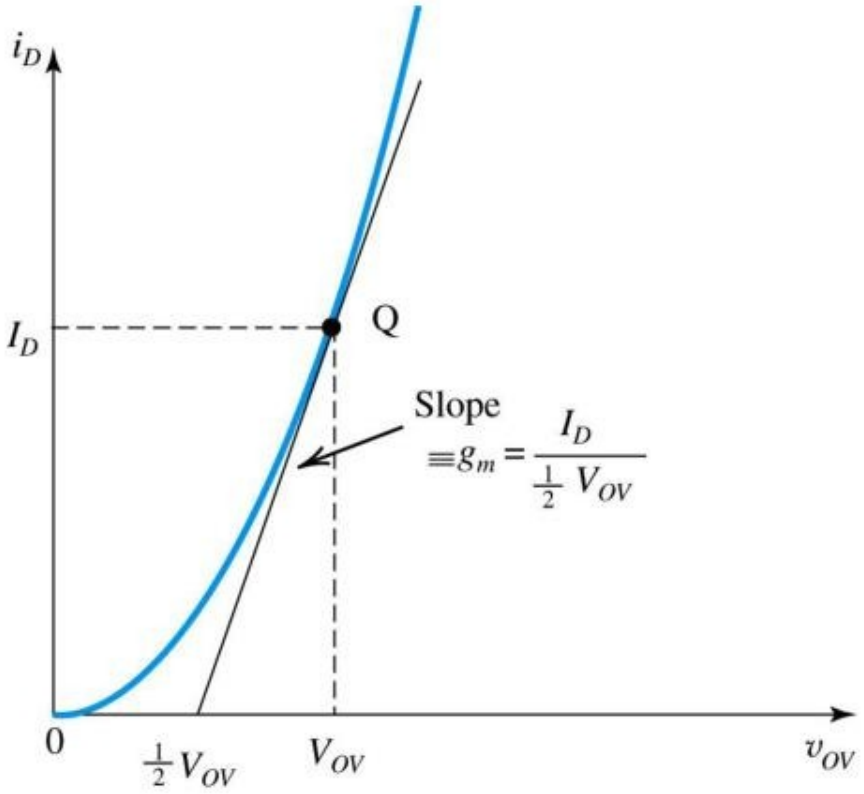


Figure 5.38: The slope of the tangent at the bias point Q intersects the v_{OV} axis at $1/2V_{OV}$. Thus $g_m = I_D/(1/2V_{OV})$.

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5.5.6: **The Transconductance g_m**

- In summary, there are three relationships for determining g_m:
  - (5.55), (5.56), and (5.57)
- These relationships are dependent on three design parameters:
  - W/L, V_{OV}, I_D

(eq5.55) $$g_m = k_n' \frac{W}{L} V_{OV}$$

(eq5.56) $$g_m = \sqrt{2k_n'} \sqrt{W/L} \sqrt{I_D}$$

(eq5.57) $$g_m = \frac{2I_D}{V_{OV}}$$

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Example 5.10: MOSFET Amplifier

- **Example 5.10 Problem Statement:** Figure 5.39(a) shows a discrete common-source MOSFET amplifier utilizing a drain-to-gate resistance R_G for biasing purposes. Such a biasing arrangement will be studied in Section 5.7. The input signal v_I is coupled to the gate via a large capacitor, and the output signal at the drain is couppled to the load resistance R_L via another large capacitor. The transistor has $V_t = 1.5V$, $k'_n(W/L) = 0.25mA/V^2$, and $V_A = 50V$. Assume the coupling capacitors to be sufficiently large so as to act as short circuits at the signal-frequencies of interest.
- **Q:** We wish to analyze this amplifier circuit to determine its **(a)** small-signal voltage gain, its **(b)** input resistance, and the largest allowable input signal.

note: capacitors block dc signals completely, but have no effect on small-signal

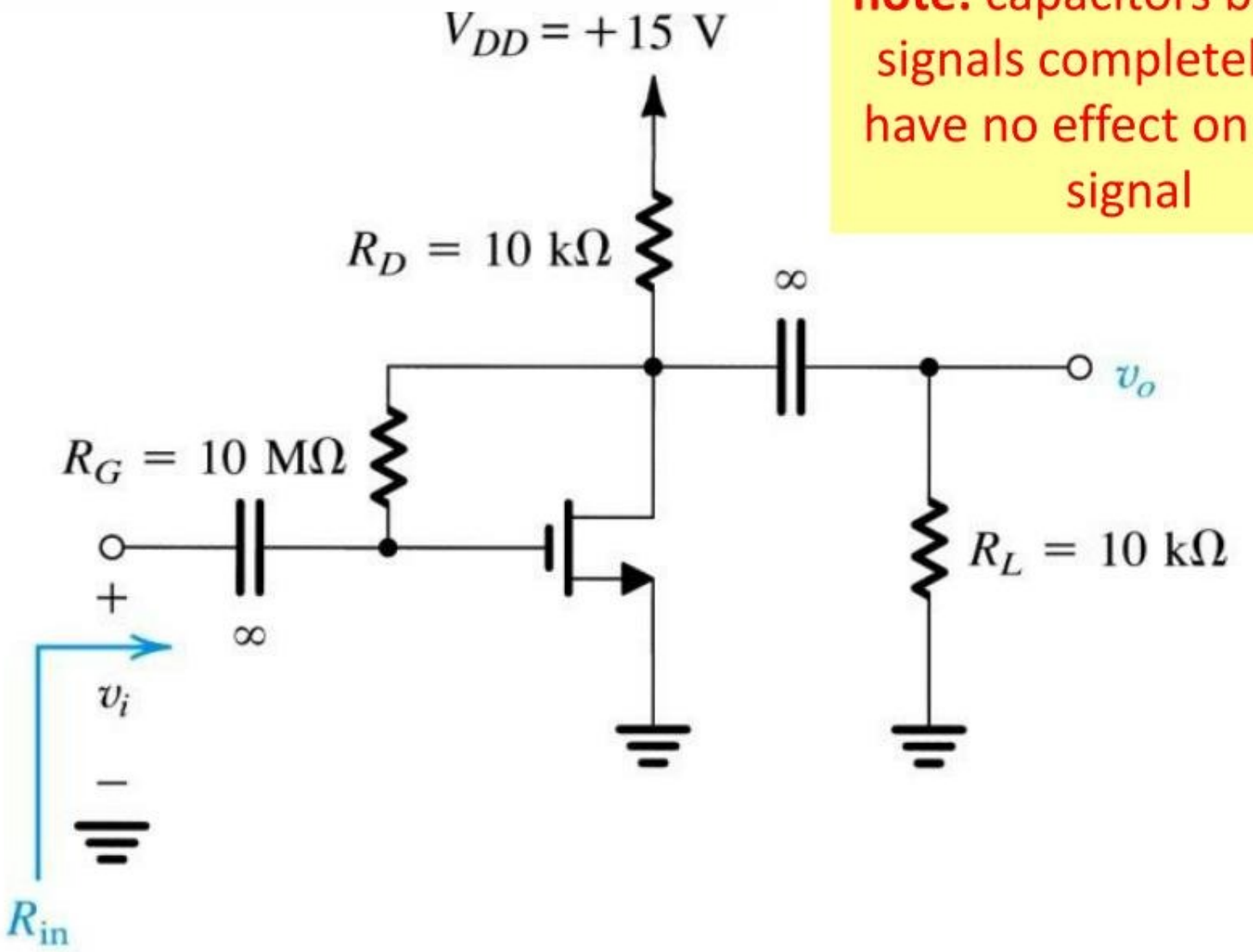


Figure 5.39: Example 5.10 amplifier circuit.

5.5.7. The T Equivalent-Circuit Model

- Through circuit transformation, it is possible to develop alternative circuit models
- ***T*-Equivalent-Ckt Model** is shown to right.

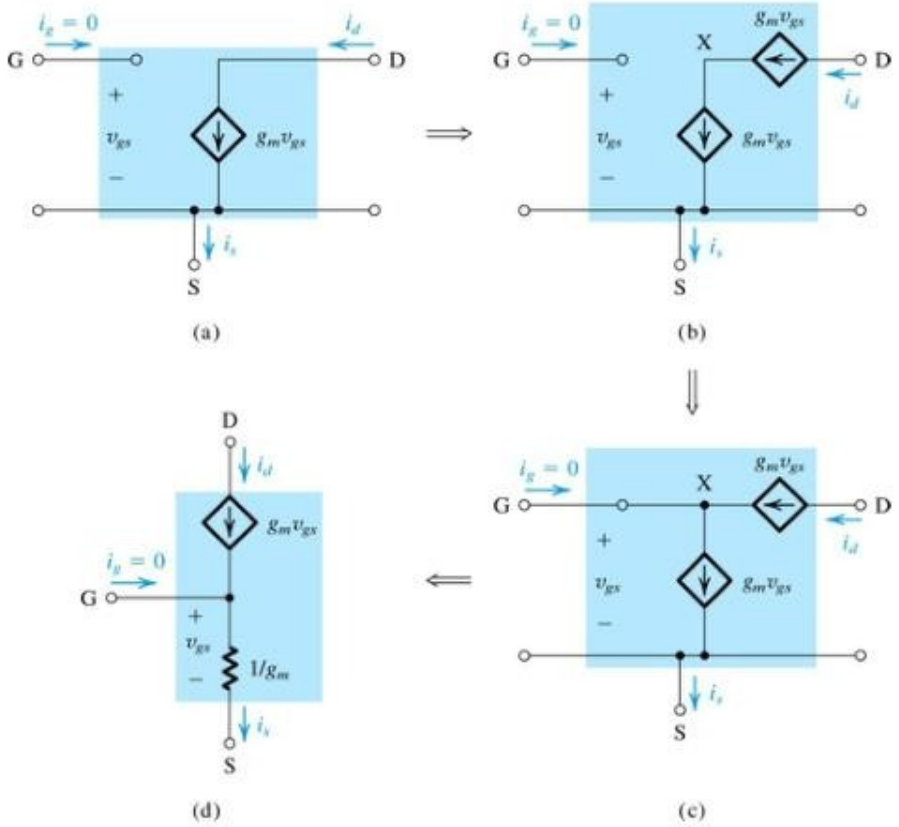


Figure 5.40: Development of the T equivalent-circuit model for the MOSFET. For simplicity, r_o has been omitted; however, it may be added between D and S in the T model of **(d)**.

5.5.7. The T Equivalent-Circuit Model

- **Q:** How is this model developed?
  - **step #1:** Begin with **small signal model** (assume R_o=0).
  - **step #2:** Place second current source in series with the first.
    - Has no effect on circuit operation.

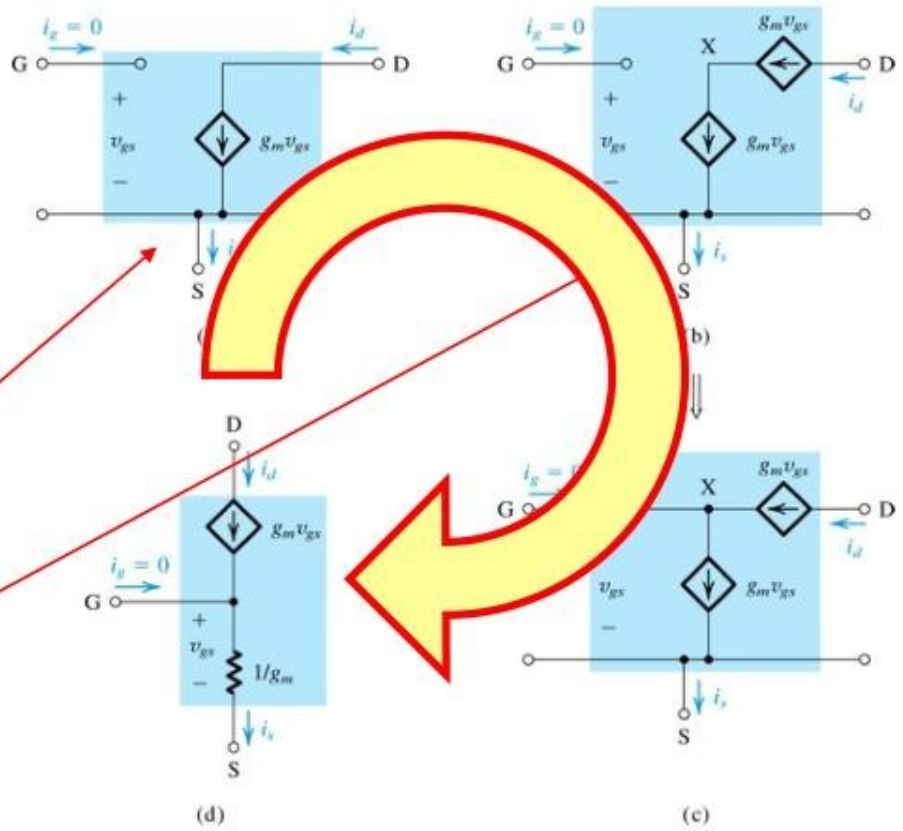


Figure 5.40: Development of the T equivalent-circuit model for the MOSFET. For simplicity, r_o has been omitted; however, it may be added between D and S in the T model of **(d)**.

Q: How is T Equivalent-Circuit Model developed?

- **step #3:** Create new node *X*, which connects gate and drain terminals
  - b/c the two current sources are equal, $i_g = 0$
- **step #4:** replace initial current source with equivalent resistance.
  - $i_{DS} = g_m v_{gs} = v_{gs}/R_{gs}$

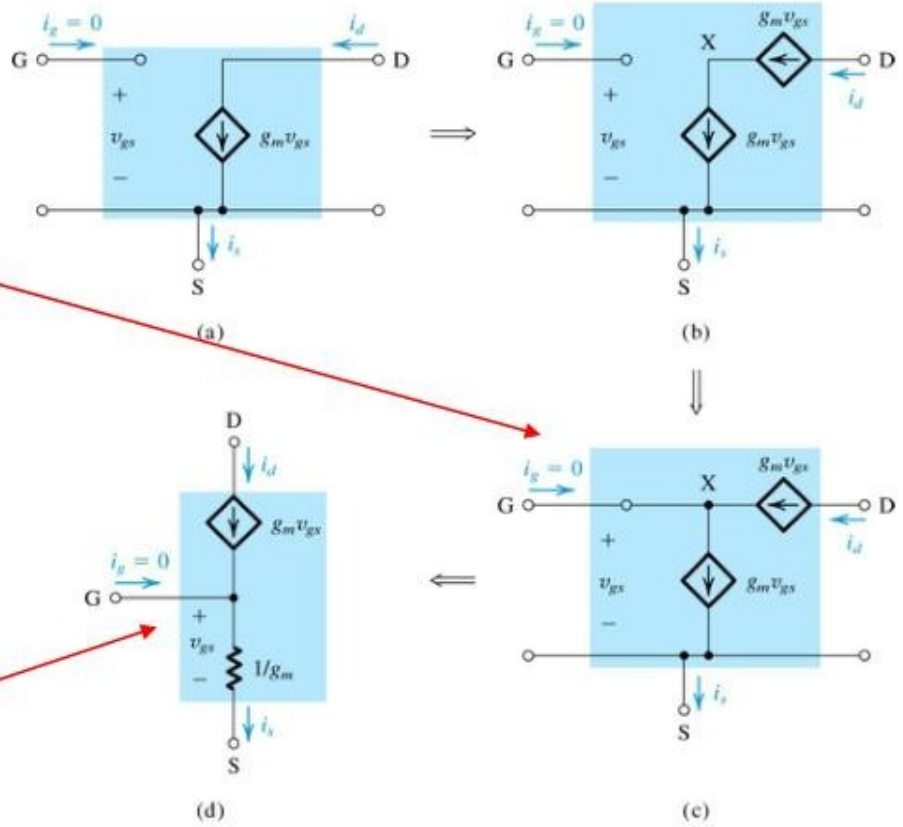


Figure 5.40: Development of the T equivalent-circuit model for the MOSFET. For simplicity, r_o has been omitted; however, it may be added between D and S in the T model of **(d)**.

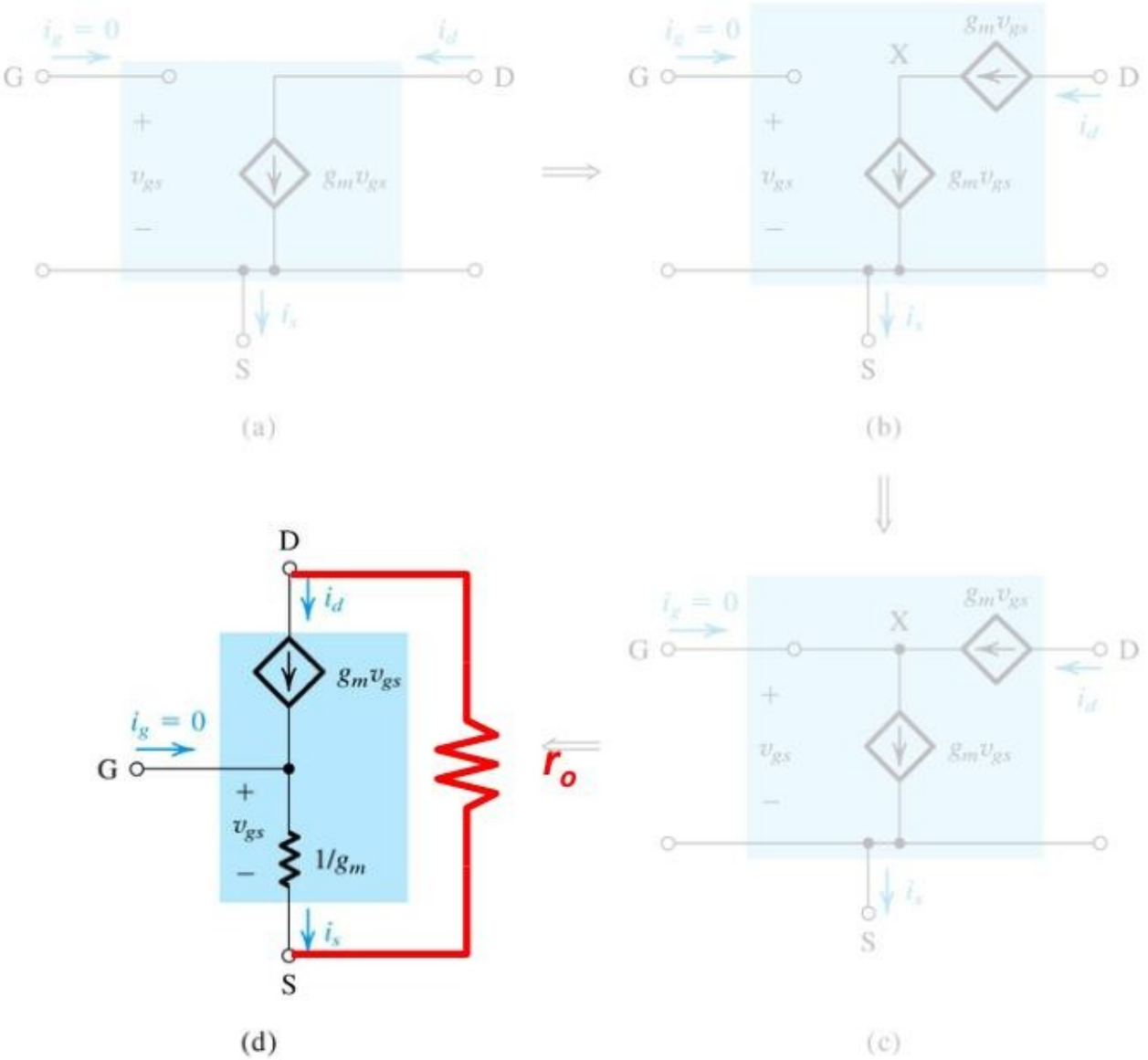


Figure 5.40: Development of the T equivalent-circuit model for the MOSFET. For simplicity, r_o has been omitted; however, it may be added.

Table 5.3 Small-Signal Equivalent-Circuit Models for the MOSFET

Small-Signal Parameters

NMOS transistors

- Transconductance:

$$g_m = \mu_n C_{ox}\frac{W}{L}V_{OV} = \sqrt{2\mu_n C_{ox}\frac{W}{L}I_D} = \frac{2I_D}{V_{OV}}$$

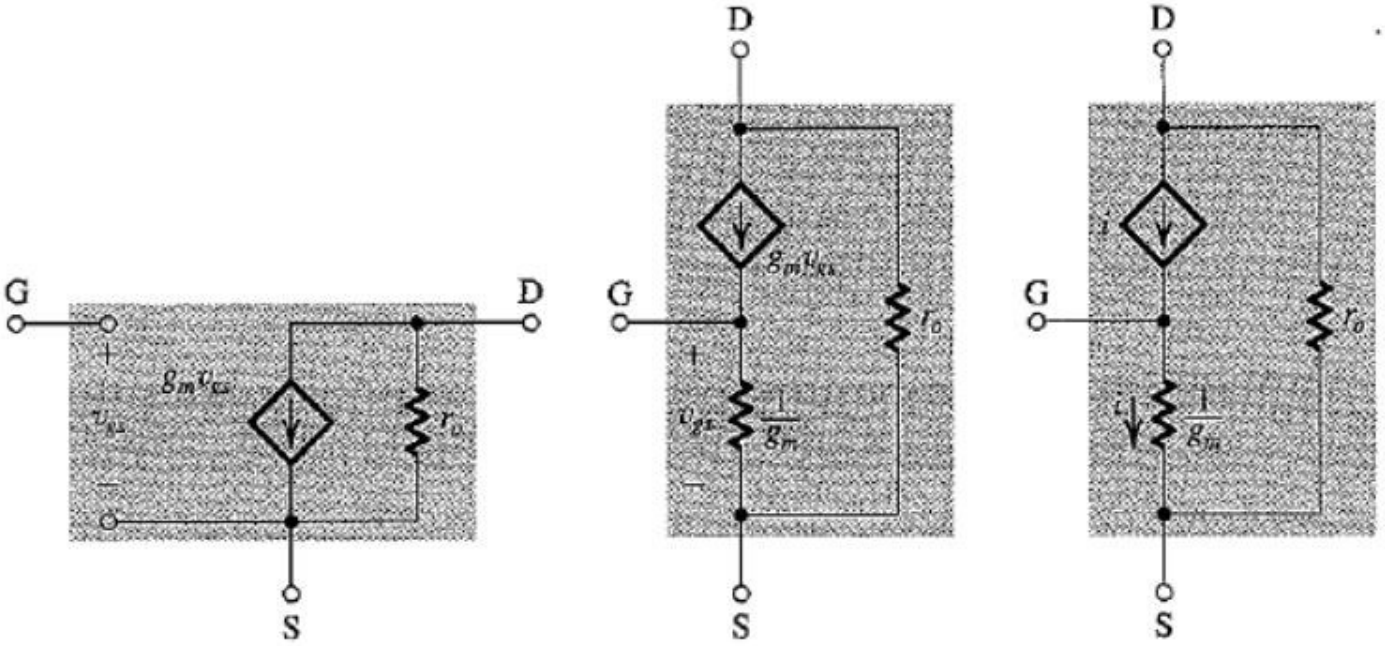
- Output resistance:

$$r_o = V_A/I_D = 1/\lambda I_D$$

PMOS transistors

Same formulas as for NMOS *except* using $|V_{ov}|$, $|V_A|$, and replacing μ_n with μ_p.

Small-Signal Equivalent Circuit Models



Hybrid-π model

T models

Summary

- The enhancement-type MOSFET is current the modt widely used semiconductor device. It is the basis of CMOS technology, which is the most popular IC fabrication technology at this time. CMOS provides both n-channel (NMOS) and p-channel (PMOS) transistors, which increases design flexibility. The minimum MOSFET channel length achievable with a given CMOS process is used to characterize the process
- The overdrive voltage $|V_{OV}| = |V_{GS}| - |V_t|$ is the key quantity that governs the operation of the MOSFET. For amplifier applications, the MOSFET must operate in the saturation region.

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Summary

- In saturation, i_D shows some linear dependence on v_{DS} as a result of the change in channel length. This channel-length modulation phenomenon becomes more pronounced as L decreases. It is modeled by ascribing an output resistance $r_o = |V_A|/I_D$ to the MOSFET model. Although the effect of ro on the operation of discrete-circuit MOS amplifiers is small, that is not the case in IC amplifiers.
- The essence of the use of MOSFET as an amplifier is that in saturation v_{GS} controls i_D in the manner of a voltage-controller current source. When the device is dc biased in the saturation region, a small-signal input (v_{gs}) may be amplified linearly.

Summary

- In cases where a resistance is connected in series with the source lead of the MOSFET, the T model is the most conveinant to use.
- The three basic configurations of the MOS amplifiers are shown in Figure 5.43.
- The CS amplifier has an ideally infinite input resistance and reasonably high gain – but a rather high output resistance and limited frequency response. It is used to obtain most of the gain in a cascade amplifier.
- Adding a resistance R_s in the source lead of the CS amplifier can lead to beneficial results.

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Summary

- The CG amplifier has a low input resistance and thus it alone has limited and specialized applications. However, its excellent high-frequency response makes it attractive in combination with the CS amplifier.
- The source follow has (ideally) infinite input resistance, a voltage gain lower than but close to unity, and a low output resistance. It is employed as a voltage buffer and as the output stage of a multistage amplifier.
- A key step in the design of transistor amplifiers is to bias the transistor to operate at an appropriate point in the saturation region.